Aerospace Structural Design: Design, Manufacturing, Testing, and Analysis of Compressive Failure on Stiffened Panels

This article presents an integrated methodology for evaluating the compressive behavior of stiffened aluminum panels through coordinated design, analysis, and validation framework. The study examines panels with C-stringer and Hat-stringer configurations and emphasizes how manufacturing practices such as forming, drilling, and fastening influence structural reliability. The theoretical component applies classical structural design principles and incorporates an optimization routine to refine geometric parameters based on predicted compressive performance. Analytical assessment is conducted using finite element modeling to capture detailed structural responses beyond what is attainable with closed-form methods. An experimental setup provides the physical validation needed to confirm analytical trends and establish the reliability of the proposed approach. Together, these components form a structured methodology for linking design decisions, fabrication quality, and predictive models, supporting the development of improved assessment practices for lightweight aerospace panel structures.

Nomenclature

A	=	Area
A_{eff}	=	Total Effective Area
A_{st}	=	Area of Stringer
A_{tot}	=	Total Area
b	=	Length
b_f	=	Length of Flange
b_s	=	Length of Sheet
b_w	=	Length of Web
c	=	End Fixity Coefficient or Rivet Head Type
E	=	Young's Modulus of Elasticity
E_t	=	Tangent Modulus of Elasticity
f	=	Effective Rivet Offset
g	=	Number of Cuts and Flanges on a Structure
h	=	Height
I	=	Moment of Inertia
k_s	=	Shear Buckling Coefficient
k_w	=	Wrinkling Coefficient
L_e	=	Effective Length
lbf	=	Pound Force
n	=	Parameter that Describes the Shape of the Stress-Strain Curve in the Yield Region
P	=	Rivet Spacing
P_c	=	Column Failing Stress
P_F	=	Panel Failing Load
$(P_C)_{ST1}$	=	Column Failing Stress of Stringer 1
ρ	=	Radius of Gyration
S	=	Distance from centerline of sheet to neutral axis of stiffener
t_f	=	Thickness of Flange
t_s	=	Thickness of Sheet
t_w	=	Thickness of Web

ν	=	Poisson's Ratio
w	=	Sheet Effective Width
w_{eff}	=	Total Effective Sheet Width
$\sigma_{0.7}$	=	Stress at the Point Where the Secant Lines at Slope 0.7E
σ_c	=	Column Failing Stress
$(\sigma_c)_{ST1}$	=	Column Failing Strength of One Stringer
σ_{cs}	=	Crippling Stress
$(\sigma_{cs})_M$	=	Monolithic Crippling Stress
$(\sigma_{cs})_{ST1}$	=	Crippling Stress of One Stringer
σ_{cr}	=	Critical Stress
σ_{cy}	=	Compression Yield Stress
σ_F	=	Panel Failing Stress
σ_{ir}	=	Inter-Rivet Buckling Stress
$(\sigma_{ir})_M$	=	Monolithic Inter-Rivet Buckling Stress
$(\sigma_{ir})_{ST1}$	=	Inter-Rivet Buckling Stress of One Stringer
σ_w	=	Sheet Wrinkling Stress
η	=	Plasticity Reduction Factor

I. Introduction

A. Objectives

The primary objective of this structural design project was to assess the performance of two stiffener geometries, namely Hat and C stringers. This was done through a combination of theoretical analysis, analytical evaluation, and experimental testing. This effort began with the validation of structural theory by calculating the buckling stresses, crippling stresses, and predicted failure loads for each configuration. Demonstrating consistency among the theoretical, analytical, and experimental results was essential to confirming the reliability of the structural models used to characterize failure behavior.

A secondary objective was to develop proficiency in the fabrication of aluminum skin stringer assemblies. Four test panels, consisting of two Hat-stringer and two C-stringer configurations, were manufactured through precise cutting, bending, drilling, and riveting operations. Following fabrication, each panel underwent routing, potting, and controlled compression testing to failure. These procedures enabled direct observation of local buckling and provided quantitative measurements of the ultimate loads sustained by each design.

The final objective was to systematically compare the performance and manufacturability of the two stiffener types. By evaluating discrepancies between predicted and experimentally measured failure loads, the project aimed to assess the accuracy of the analytical methods used. It also examined the buckling and failure characteristics of each panel to understand the structural behavior of the two geometries. Additionally, differences in fabrication time and manufacturing complexity were considered to provide a practical comparison. Together, these assessments were used to determine the relative efficiency of the Hat and C stringer configurations.

B. Schedule

The project timeline began with the design and manufacturing phase. Work commenced immediately in early September with the completion of the design phase, which involved developing detailed models of all components in SolidWorks. Following this, the project transitioned into the more labor-intensive manufacturing period. This stage included generating detailed engineering, cutting the stringers to specifications, bending both the C and Hat stringer geometries, and performing approximately seven hours of precision drilling and riveting to assemble the skin stringer panel. This phase was completed by mid-September and was essential to ensure that the test panels met the required dimensional tolerances for reliable evaluation.

The second half of the project focused on test preparation and analysis. Throughout October, efforts were directed toward preparing the specimens for structural testing. This included several days dedicated to wood routing and potting the ends of each panel to ensure proper load transfer during compression testing. The primary experimental setup and validation work was conducted on October 28th, during which the test configuration was finalized and verified. The final stage of the project, conducted during November, centered on analytical evaluation. Approximately twelve hours were spent comparing theoretical predictions with the experimental data obtained during testing. This was followed by additional work on design optimization and an extensive Abaqus modeling effort, totaling approximately fifteen hours, to generate the numerical results required for comprehensive comparison and final conclusions.

II. Design and Manufacturing

A. Panel Designs

The panel design incorporated two distinct stiffener configurations: each selected for their unique structural characteristics and load carrying capabilities. The C stringer is a streamlined structural element intended to provide localized reinforcement while efficiently transferring shear and compression loads through the panel. Linear pattern of fastener holes facilitates straightforward assembly and offers a predictable load path along the length of the stiffener. In contrast, the Hat stringer is formed with a more complex geometry that enhances both stiffness and overall structural stability. This configuration is particularly effective in distributing shear loads due to its increased moment of inertia and the presence of multiple fastener locations, which improve load sharing between the stringer and the skin. To support these stiffeners, two corresponding panel plates were used. The C panel plate is a thin, flat component designed to reinforce the regions where fasteners interface with the C stringer. This added reinforcement enhances the panel’s ability to withstand shear loads and reduces the likelihood of localized deformation. Similarly, the Hat panel plate provides a strengthened attachment surface for the Hat stringer. By distributing loads more effectively, it ensures proper shear transfer and helps maintain structural integrity under applied loading. Together, these components create a unified panel system capable of carrying the required loads and supporting a clear comparison of the two stiffener geometries.

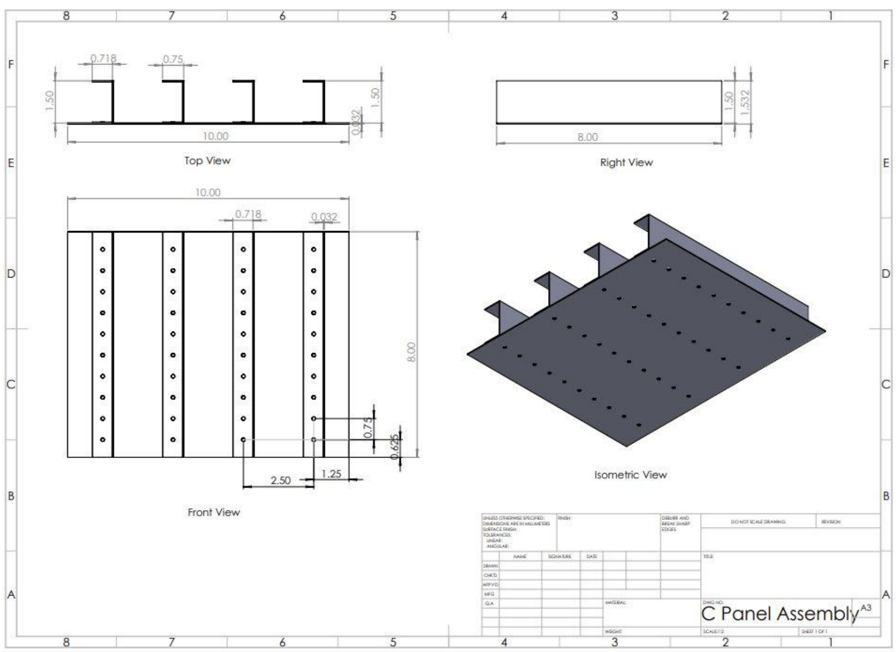

Figure 1. SolidWorks Drawing for C Panel Assembly

Figure 2. SolidWorks Drawing for Hat Panel Assembly

B. Manufacturing Process

The manufacturing of the test panels was carried out through a structured five step process that emphasized precision and consistency at every stage. The first step involved creating accurate drawing layouts by transferring all required dimensions from the engineering blueprints onto the aluminum sheet metal. Calipers and straight edges were used to ensure dimensional accuracy, and each measurement was verified through a two-person process in which one team member marked the layout while the other independently confirmed it. This collaborative approach minimized the accumulation of measurement error and ensured a reliable foundation for subsequent manufacturing steps. The drawing layouts step is shown below in Figure 3. The second step consisted of cutting the sheet metal using a manual foot shear. To maintain dimensional control, cuts were intentionally made with a small margin of excess material, allowing for later refinement. Alignment was achieved by matching the drawn line with the shear blade and using the integrated guide square on the shear to maintain straight and consistent cuts. The metal cutting step is shown below in Figure 4.

The third phase of manufacturing involved bending the stringers to form both the C and Hat profiles. A manual sheet metal brake was used for this operation, with particular attention given to compensating for material spring back. Because the aluminum tended to return to a slightly smaller angle after bending, the material was deliberately over bent and immediately checked with a protractor to ensure a final bend angle of 90°. The metal bending step is shown below in Figure 5. Following bending, precision drilling was performed using a bench mounted drill press to create the rivet holes. A straight metal guide was clamped to the drill table to preserve alignment along the fastener line, and each hole locating was visually checked prior to drilling to avoid misalignment caused by minor part movement. The precision drilling step is shown below in Figure 6.

The final step in the process was the riveting and assembly of the skin stringer panels. A handheld rivet tool was used to install the fasteners. During early assembly, small gaps were observed between the components, which were traced to residual metal shavings left from drilling. To prevent this issue, all components were thoroughly cleaned prior to riveting, and a standardized procedure was established to clean parts immediately after drilling. Rivets were installed beginning at the four corners and progressing inward, a method that ensured uniform clamping pressure and resulted in a structurally sound and well aligned joint. The riveting step is shown below in Figure 7.

Figure 3. Example of Drawing Layouts

Figure 4. Example of Metal Cutting

Figure 5. Example of Stringer Bending

Figure 6. Example of Precision Drilling

Figure 7. Example of Riveting

C. Manufactured Panels

Upon completion of the manufacturing process, four test panels were produced. The C panel assemblies consisted of four C-stringers riveted to a single C panel plate, forming a multi-channel reinforced structure. This configuration provided a straightforward yet structurally effective geometry for channel type reinforcement. In contrast, the Hat panel assemblies incorporated two wide Hat stringers affixed to a Hat panel plate. Due to their cross-sectional profile, the Hat stringers offered increased stiffness and enhanced stability under compressive loading compared to the C-panel design. Following assembly, all panels underwent a thorough visual inspection to verify proper installation, confirm geometric consistency, and ensure overall structural quality. This inspection validated the precision and workmanship achieved during fabrication. Displayed in Figure 8, the four completed and verified panels were prepared for subsequent compression load testing at the Advanced Composite Institute (ACI).

Figure 8. All Panels After the Completion of Manufacturing

D. Learning Curve Analysis

To evaluate efficiency improvements achieved during the manufacturing and test preparation phases, a learning curve analysis was conducted using the standard in class structural learning curve model. The analysis recorded the time required to complete the five primary tasks, namely the drawing layouts, cutting metal, bending stringers, drilling rivets holes, and riveting. Across all tasks, a notable reduction in time per unit was observed, particularly during the early repetitions, indicating rapid proficiency gains and effective acquisition of the necessary manual skills. These results were reflected in the overall panel manufacturing and test preparation learning curves, which clearly illustrated the downward trend in task duration. Shown in Figure 9, although only four test panels were fabricated, the learning curves were extrapolated to ten units to estimate potential long term production efficiency and associated labor cost reductions in a scaled manufacturing environment. The analysis provides clear evidence of the team's capacity to refine production techniques, streamline workflow, and reduce manufacturing time through iterative practice and process optimization.

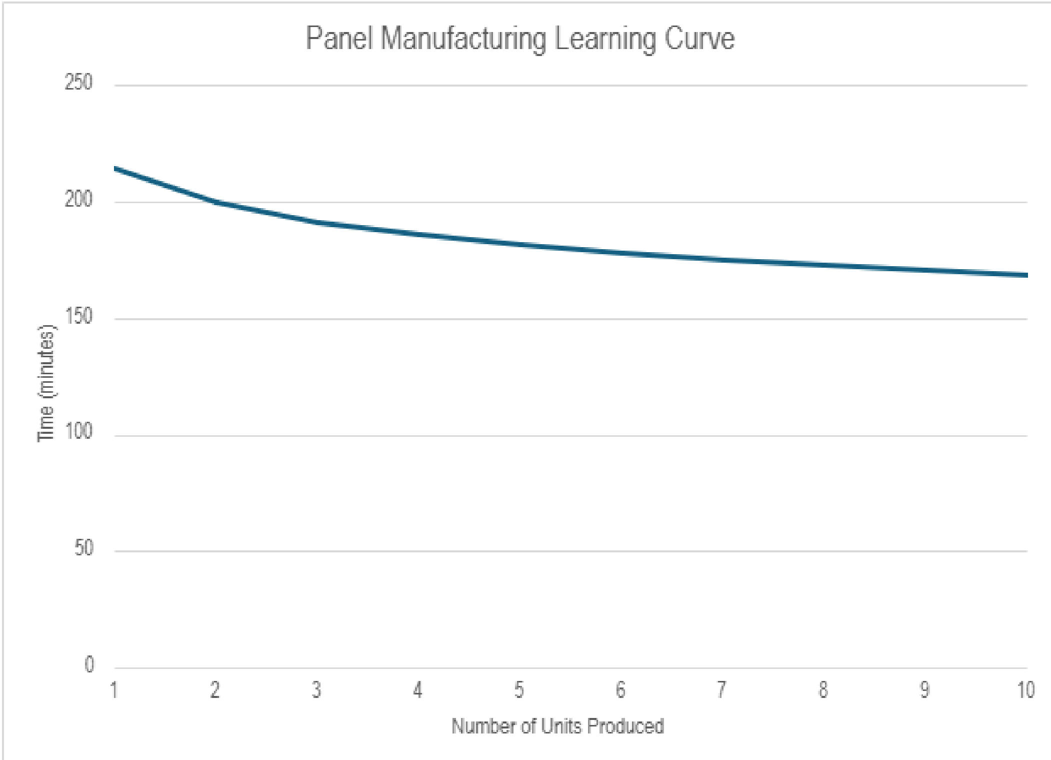

Figure 9. Panel Manufacturing Learning Curve

III. Theoretical Analysis & Optimization

A. Theoretical Backgrounds

The theoretical phase of this project focused on establishing the maximum load limits associated with compressive failure. Fundamental principles from aerospace structural mechanics were applied to predict the ultimate strength of both the C-stringer and Hat-stringer configurations. The objective was to determine the failing stresses and corresponding loads for each stringer type by evaluating multiple forms of buckling and crippling behavior through established stress and load formulations.

Local elastic buckling stresses were calculated for the unsupported thin elements of each stringer, including the webs and flanges. Classical plate buckling theory was applied and buckling coefficients were selected according to the boundary conditions of each element. Because aluminum displays nonlinear behavior as it approaches the yield region, the Ramberg-Osgood relation was used to incorporate plastic corrections and preserve accuracy in the nonlinear stress-strain range. Crippling stresses were then computed to establish the maximum load each cross section could sustain before collapse through buckling and wrinkling. These values were obtained using monolithic crippling stress expressions and multi-corner section equations. Both approaches account for the number of cuts and flanges as well as the effective width within the cross-section. This allowed for the calculation of the ultimate theoretical material capacity associated with crippling behavior to be studied.

By combining these buckling and crippling calculations, the theoretical analysis produced a baseline estimate of the minimum expected load capacity for each geometry. These predictions formed the foundation for the subsequent stages of the project. They served as the reference against analytical simulations in Abaqus and the experimental data from physical testing. This enabled a structured evaluation of analytical accuracy and validation of the manufactured panels' performance.

B. Theoretical Analysis of C-Stringer and C-Stiffened Panel

Part 1. Determine the Local Buckling Stress of the C-Stringer

1. Find k_w

$$b_w = h - 2\left(\frac{t_f}{2}\right) = 1.5\ in - 2\left(\frac{0.032\ in}{2}\right) = 1.468\ in$$

$$b_f = b - \frac{t_w}{2} = 0.75\ in - \frac{0.032\ in}{2} = 0.734\ in$$

$$t_w = t_f = 0.032\ in$$

$$\frac{b_f}{b_w} = \frac{0.734\ in}{1.468\ in} = 0.5\ ;\ \frac{t_w}{t_f} = \frac{0.032\ in}{0.032\ in} = 1$$

Using **Figure C6.4**: $k_w = 2.9$

2. Find the Elastic Buckling Stress

$$(\sigma_{cr})_e = \frac{k_w \pi^2 E}{12(1-\nu^2)} * \left(\frac{t_w}{b_w}\right)^2 = \frac{(2.9)(\pi^2)(10.7\ x10^6 psi)}{12(1-(0.3)^2)} * \left(\frac{0.032\ in}{1.468\ in}\right)^2 = 13{,}626.212\ psi$$

3. Check Elasticity with Ramberg-Osgood Equation

$$\frac{E_t}{E} = \frac{1}{1 + \frac{3}{7} n \left(\frac{\sigma}{\sigma_{0.7}}\right)^{n-1}} = \frac{1}{1 + \frac{3}{7}(11.5)\left(\frac{13{,}626.212\ psi}{39{,}000\ psi}\right)^{11.5-1}} = 0.9999 < 1$$

4. Find Local Buckling Stress with Plastic Correction

$$\frac{k_w \pi^2 E}{12(1-\nu^2)\sigma_{0.7}} * \left(\frac{t_w}{b_w}\right)^2 = \frac{(2.9)(\pi^2)(10.7\ x10^6 psi)}{12(1-(0.3)^2)(39{,}000\ psi)} * \left(\frac{0.032\ in}{1.468\ in}\right)^2 = 0.3417$$

Using **Figure C5.7**: $\frac{\sigma_{cr}}{\sigma_{0.7}} = \eta = 0.345$

$$(\sigma_{cr})_p = \eta \sigma_{0.7} = (0.345)(39{,}000\ psi) = 13{,}455\ psi$$

Local Buckling Stress of C-Stringer: $\boldsymbol{(\sigma_{cr})_p = 13{,}455\ psi}$

Part 2. Determine the Local Buckling Stress of the C-Stiffened Panel

1. Find k_s

$$\frac{b_f}{b_w} = \frac{0.734\ in}{1.468\ in} = 0.5\ ;\ \frac{t_w}{t_f} = \frac{0.032\ in}{0.064\ in} = 0.5\ ;\ \frac{b_w}{b_s} = \frac{1.468\ in}{2.5\ in} = 0.5872$$

Using **Figure C6.9**: $k_s = 2.6$

2. Find the Elastic Buckling Stress

$$(\sigma_{cr})_e = \frac{k_s \pi^2 E}{12(1-\nu^2)} * \left(\frac{t_s}{b_s}\right)^2 = \frac{(2.6)(\pi^2)(10.7\ x10^6 psi)}{12(1-(0.3)^2} * \left(\frac{0.064\ in}{2.5\ in}\right)^2 = 16{,}478.36\ psi$$

3. Check Elasticity with Ramberg-Osgood Equation

$$\frac{E_t}{E} = \frac{1}{1+\frac{3}{7}n\left(\frac{\sigma}{\sigma_{0.7}}\right)^{n-1}} = \frac{1}{1+\frac{3}{7}(11.5)\left(\frac{16{,}478.36\ psi}{39{,}000\ psi}\right)^{11.5-1}} = 0.9994 < 1$$

4. Find Local Buckling Stress with Plastic Correction

$$\frac{k_w \pi^2 E}{12(1-\nu^2)\sigma_{0.7}} * \left(\frac{t_s}{b_s}\right)^2 = \frac{(2.6)(\pi^2)(10.7\ x10^6 psi)}{12(1-(0.3)^2)(39{,}000\ psi)} * \left(\frac{0.064\ in}{2.5\ in}\right)^2 = 0.4225$$

Using **Figure C5.8**: $\frac{\sigma_{cr}}{\sigma_{0.7}} = \eta = 0.425$

$$(\sigma_{cr})_p = \eta\sigma_{0.7} = (0.425)(39{,}000\ psi) = 16{,}575\ psi$$

<u>Local Buckling Stress of C-Stiffened Panel</u>: $\boldsymbol{(\sigma_{cr})_p = 16{,}575\ psi}$

Part 3. Determine the Monolithic Crippling Stress

$$\frac{(\sigma_{cs})_M}{\sigma_{cy}} = 0.56\left[\left(\frac{gt^2}{A}\right)\left(\frac{E}{\sigma_{cy}}\right)^{\frac{1}{2}}\right]^{0.85} = 0.56\left[\left(\frac{(8)(0.032\ in)^2}{(0.093952\ in^2)}\right)\left(\frac{10.7\ x10^6 psi}{(39{,}000\ psi)}\right)^{\frac{1}{2}}\right]^{0.85} = 0.4244 < 0.8$$

$$\frac{(\sigma_{cs})_M}{\sigma_{cy}} = 0.4244$$

$$(\sigma_{cs})_M = 0.4244\sigma_{cy} = 0.4244(39{,}000\ psi) = 16{,}969\ psi$$

<u>Monolithic Crippling Stress</u>: $\boldsymbol{(\sigma_{cs})_M = 16{,}969\ psi}$

Part 4. Determine the Monolithic Inter-Rivet Buckling Stress

$$\frac{c\pi^2 E}{12(1-\nu^2)\sigma_{0.7}}\left(\frac{t_s}{P}\right)^2 = \frac{(3)(\pi^2)(10.7\ x10^6 psi)}{12(1-(0.3)^2)(39{,}000\ psi)}\left(\frac{0.064\ in}{0.75\ in}\right)^2 = 1.2656$$

Using **Figure C7.24**: $\frac{\sigma_{ir}}{\sigma_{0.7}} = 0.91$

$$(\sigma_{ir})_M = 0.91\sigma_{0.7} = 0.91(39{,}000\ psi) = 35{,}490\ psi$$

<u>Monolithic Inter-Rivet Buckling Stress</u>: $\boldsymbol{(\sigma_{ir})_M = 35{,}490\ psi}$

Part 5. Determine the Sheet Wrinkling Stress

1. Find f

$$b_0 = \frac{b}{2} - \frac{t_w}{2} = \frac{0.75\ in}{2} - \frac{0.032\ in}{2} = 0.359\ in$$

$$\frac{b_0}{t_w} = \frac{0.359\ in}{0.032\ in} = 11.2187; \quad \frac{p}{d} = \frac{0.75\ in}{.125\ in} = 6$$

Using **Figure 7**: $\frac{f}{t_w} = 9.8$

$$\rightarrow f = 9.8(t_w) = 9.8(0.032\ in)$$

$$\rightarrow f = 0.3136\ in$$

2. Find k_w

$$b_w = h - t_s = 1.5\ in - 0.032\ in = 1.468\ in$$

$$\frac{b_w}{t_w} = \frac{1.468\ in}{0.032\ in} = 45.875; \quad \frac{b_s}{t_s} = \frac{2.5\ in}{0.032\ in} = 78.125; \quad \frac{b_w/t_w}{b_s/t_s} = \frac{45.875}{78.125} = 0.5872$$

$$\frac{f}{b_w} = \frac{0.3136\ in}{1.468\ in} = 0.21362$$

Using **Figure 2:** $k_w = 10$

3. Find σ_w

$$\frac{k_w \pi^2 E}{12(1-\nu^2)\sigma_{0.7}} * \left(\frac{t_s}{b_s}\right)^2 = \frac{(10)(\pi^2)(10.7 * 10^6 psi)}{12(1-(0.3)^2)(39{,}000\ psi)} * \left(\frac{0.032\ in}{2.5\ in}\right)^2 = 0.37969$$

Using **Figure C7.24:** $\frac{\sigma_w}{\sigma_{0.7}} = 0.375 < 0.9 \rightarrow OK!$

$$\rightarrow \sigma_w = 0.375(\sigma_{0.7}) = 0.375(39{,}000\ psi) = 14{,}625\ psi$$

Sheet Wrinkling Stress: $\boldsymbol{\sigma_w = 14{,}625\ psi}$

Part 6. Determine the Crippling Stress of One C-Stringer

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 3.2\left[\left(\frac{t^2}{A}\right)\left(\frac{E}{\sigma_{cy}}\right)^{\frac{1}{3}}\right]^{0.75} = 3.2\left[\left(\frac{(0.032\ in)^2}{(0.093952\ in^2)}\right)\left(\frac{(10.7\ x10^6 psi)}{(39{,}000\ psi)}\right)^{\frac{1}{3}}\right]^{0.75} = 0.4292 < 0.9$$

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.4292$$

$$\sigma_{cs} = 0.4292\sigma_{cy} = 0.4292(39{,}000\ psi) = 17{,}168\ psi$$

Crippling Stress of One C-Stringer: $\boldsymbol{(\sigma_{cs})_{ST1} = 17{,}168\ psi}$

Part 7. Determine the Column Failing Stress of One C-Stringer

1. Find the Effective Length

$$L_e = \frac{L}{\sqrt{c}} = \frac{8\ in}{\sqrt{4}} = 4\ in$$

2. Find the Radius of Gyration

$$\rho_x = \sqrt{\frac{I_X}{\sum A_i}} = \sqrt{\frac{0.033760834\ in^4}{0.093952\ in^2}} = 0.5995\ in$$

3. Find the Slenderness Ratio

$$\frac{L_e}{\rho_x} = \frac{4\ in}{0.5995\ in} = 6.6722 < 20$$

4. Find the Column Failing Stress of One C-Stringer

$$\sigma_c = \sigma_{cs} - \frac{(\sigma_{cs})^2}{4\pi^2 E}\left(\frac{L_e}{\rho_x}\right)^2 = (17{,}168\ psi) - \frac{(17{,}168\ psi)^2}{(4)(\pi^2)(10.7\ x10^6 psi)}\left(\frac{4\ in}{0.5995\ in}\right)^2 = 17{,}135\ psi$$

<u>Column Failing Stress of One C-Stringer</u>: $(\boldsymbol{\sigma_c})_{ST1} = \mathbf{17{,}135}\ \boldsymbol{psi}$

Part 8. Determine the Column Failing Stress of One C-Stringer with Effective Width

1. Find the Effective Width

$$W_0 = 1.9\left(t\sqrt{\frac{E}{\sigma_{cs}}}\right) = 1.9\left((0.032\ in)\sqrt{\frac{(10.7\ x10^6 psi)}{(17{,}168\ psi)}}\right) = 1.4688\ in < 2.5 in$$

$$W_1 = 0.62\left(t\sqrt{\frac{E}{\sigma_{cs}}}\right) = 0.62\left((0.032\ in)\sqrt{\frac{(10.7\ x10^6 psi)}{(17{,}168\ psi)}}\right) = 0.4793\ \text{in} < 0.75\ \text{in}$$

$$W_{eff}^{(0)} = W_1 + \frac{W_0}{2} = 0.4793\ in + \frac{1.4688\ in}{2} = 1.2137\ in$$

2. Find the Column Failing Stress

$$\left(\frac{\rho_1}{\rho_0}\right)^2 = \frac{1 + \left[1 + \left(\frac{S}{\rho_0}\right)^2\right]\frac{W_{eff}^{(0)} t}{A_{ST}}}{1 + \left(\frac{W_{eff}^{(0)} t}{A_{ST}}\right)^2} = \frac{1 + \left[1 + \left(\frac{\frac{t}{2} + \frac{h}{2}}{\rho_0}\right)^2\right]\frac{W_{eff}^{(0)} t}{A_{ST}}}{1 + \left(\frac{W_{eff}^{(0)} t}{A_{ST}}\right)^2}$$

$$\left(\frac{\rho_1}{\rho_0}\right)^2 = \frac{1 + \left[1 + \left(\frac{\frac{0.032\ in}{2} + \frac{1.5\ in}{2}}{(0.5995\ in)}\right)^2\right]\frac{(1.2137\ in)(0.032\ in)}{(0.093952\ in^2)}}{1 + \left(\frac{(1.2137\ in)(0.032\ in)}{(0.093952\ in^2)}\right)^2} = 1.7835$$

$$\rho_1 = \sqrt{1.7835(\rho_0)^2} = \sqrt{1.7835(0.5995\ in)^2} = 0.80057\ in$$

$$\frac{L_e}{\rho_1} = \frac{4\ in}{0.80057\ in} = 4.9964 < 20$$

$$\sigma_c = \sigma_{cs} - \frac{(\sigma_{cs})^2}{4\pi^2 E}\left(\frac{L_e}{\rho_1}\right)^2 = (17{,}168\, psi) - \frac{(17{,}168\, psi)^2}{(4)(\pi^2)(10.7\, x10^6 psi)}\left(\frac{4\ in}{0.80057\ in}\right)^2 = 17{,}150.18\, psi$$

<u>Column Failing Stress of Stringer 1 with Effective Width</u>: $(\boldsymbol{\sigma_c})_{ST1} = \mathbf{17{,}150.18}\ \boldsymbol{psi}$

Part 9. Determine the Column Failing Load of One C-Stringer with Effective Area

1. Find the Effective Area

$$A_{eff}^{(1)} = W_{eff}^{(0)} t_s = (1.2137\ in)(0.064\ in) = 0.03884\ in^2$$

2. Find the Total Area

$$A_{tot} = A_{st} + A_{eff}^{(1)} = 0.1328\ in^2$$

3. Find the Column Failing Load of One C-Stringer with Effective Area

$$(P_c)_{ST1} = (\sigma_c)_{ST1}\, A_{tot} = (17{,}150.18\, psi)(0.1328\, in^2) = 2{,}277\ lbf$$

<u>Column Failing Load of One C-Stringer with Effective Area:</u> $(\boldsymbol{P_c})_{ST1} = \mathbf{2{,}277}\ \boldsymbol{lbf}$

Part 10. Determine the Inter-Rivet Buckling Stress of One C-Stringer

1. Find the Elastic Inter-Rivet Buckling Stress of One C-Stringer

$$(\sigma_{ir})_{ST1} = \frac{\pi^2 E}{\left(\frac{P}{0.29t\sqrt{c}}\right)^2} = \frac{\pi^2(10.7\ x10^6 psi)}{\left(\frac{0.75\ in}{0.29(0.032\ in)\sqrt{3}}\right)^2} = 45{,}330.9\ psi$$

2. Check Elasticity with Ramberg Osgood Equation

$$\frac{E_t}{E} = \frac{1}{1+\frac{3}{7}n\left(\frac{\sigma}{\sigma_{0.7}}\right)^{n-1}} = \frac{1}{1+\frac{3}{7}(11.5)\left(\frac{45{,}330.9\ psi}{39{,}000\ psi}\right)^{11.5-1}} = 0.0568 < 1$$

3. Find Inter-Rivet Buckling Stress of One C-Stringer with Plastic Correction

$$\frac{\rho_1}{t} = \frac{0.80057\ in}{0.032\ in} * \frac{\sqrt{4}}{\sqrt{3}} = 28.8881$$

Using **Figure C7.18**: $(\boldsymbol{\sigma_{ir}})_{ST1} = \mathbf{32{,}000}\ \boldsymbol{psi}$

<u>Inter-Rivet Buckling Stress of One C-Stringer</u>: $(\boldsymbol{\sigma_{ir}})_{ST1} = \mathbf{32{,}000}\ \boldsymbol{psi}$

Part 11. Determine the Panel Failing Strength

$$\sigma_F = \frac{\sigma_W(b_s * t_s) + min[(\sigma_{cs})_{ST1}, (\sigma_{cs})_M,\ \sigma_W, (\sigma_{ir})_M) *\ A_{ST}}{(b_s * t_s) +\ A_{ST}}$$

$$(\sigma_{cs})_{ST1} = 17{,}168\ psi$$
$$(\sigma_{cs})_M = 16{,}969\ psi$$
$$\sigma_W = 14{,}625\ psi$$
$$(\sigma_{ir})_M = 35{,}490\ psi$$

$$\sigma_W < (\sigma_{cs})_M < (\sigma_{cs})_{ST1} < (\sigma_{ir})_M$$

$$\sigma_F = \frac{(14{,}625\ psi)(2.5\ in * 0.064\ in) + (16{,}969\ psi) * (0.093952\ in^2)}{(2.5\ in * 0.064\ in) + 0.093952\ in^2} = 15{,}084\ psi$$

<u>Panel Failing Strength</u>: $\boldsymbol{\sigma_F = 15{,}084\ psi}$

Part 12. Determine the Panel Failing Load

$$P_F = \sigma_F * A_{tot} = (15{,}084\ psi)(0.1328\ in^2) = 2{,}623\ lbf$$

<u>Panel Failing Load</u>: $\boldsymbol{P_F = 2{,}623\ lbf}$

C. Theoretical Background of Hat-Stringer and Hat-Stiffened Panel
Part 1. Determine the Global Buckling Stress and Local Buckling Stress

1. Find $(\sigma_{cr})_{global}$

$$(\sigma_{cr})_{global} = \frac{\pi^2 E}{\left(\frac{L_e}{\rho}\right)^2} \;\; ; \;\; fixed - fixed, axially\ loaded{:}\ L_e = \frac{L}{\sqrt{c}} = \frac{8\ in}{2} = 4\ in$$

$$\rho_x = \sqrt{\frac{I_x}{\sum A_i}} = \sqrt{\frac{0.5465\ in^4}{0.25189\ in^2}} = 0.8035\ in$$

$$(\sigma_{cr})_{global} = \frac{\pi^2 E}{\left(\frac{L_e}{\rho}\right)^2} = \frac{\pi^2 (10.7 * 10^6\ psi)}{\left(\frac{4\ in}{1.4729\ in}\right)^2} = 4.2616 * 10^6\ psi$$

<u>Global Buckling Stress:</u> $\boldsymbol{(\sigma_{cr})_{global} = 4.2616 * 10^6\ psi}$

2. Find k_t

$$b_t = b_2 - \frac{t_w}{2} - \frac{t_w}{2} = 2\ in - \frac{0.032\ in}{2} - \frac{0.032\ in}{2} = 1.968\ in$$

$$b_w = h - \frac{t_t}{2} - \frac{t_t}{2} = 2\ in - \frac{0.032\ in}{2} - \frac{0.032\ in}{2} = 1.968\ in$$

$$b_f = b_1 - \frac{t_w}{2} = 0.984\ in$$

$$\frac{b_f}{b_w} = \frac{0.984\ in}{1.968\ in} = 0.5 \;\; ; \;\; \frac{b_w}{b_t} = \frac{1.968\ in}{1.968\ in} = 1.0$$

Using **Figure C6.9**: $k_t = 3.25$

3. Find $\boldsymbol{(\sigma_{cr})_{local}}$

$$(\sigma_{cr})_{local} = \frac{k_s \pi^2 E}{12(1-\nu^2)} * \left(\frac{t_s}{b_t}\right)^2 = \frac{(3.25)(\pi^2)(10.7\,x10^6 psi)}{12(1-(0.3)^2)} * \left(\frac{0.032\ in}{1.968\ in}\right)^2 = 8{,}309.866\ psi$$

<u>**Local Buckling Stress**</u>: $\boldsymbol{(\sigma_{cr})_{local} = 8{,}309.866\, psi}$

Part 2. Determine the Crippling Stress of Hat-Stiffened Panel

1. Find σ_{cs}

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.56\left[\left(\frac{g * t^2}{A}\right)\left(\frac{E}{\sigma_{cy}}\right)^{\frac{1}{2}}\right]^{0.85} = 0.56\left[\left(\frac{(11)(0.032\, in)^2}{(0.25189\, in^2)}\right)\left(\frac{(10.7\, x10^6 psi)}{(40{,}000\, psi)}\right)^{\frac{1}{2}}\right]^{0.85} = 0.4292 < 0.8$$

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.4292 \rightarrow OK!$$

$$\sigma_{cs} = 0.4292(\sigma_{cy}) = 0.4292(40{,}000\, psi) = 17{,}169.99\, psi$$

<u>**Crippling Stress of One Hat-Stiffened Panel**</u>: $\boldsymbol{\sigma_{cs} = 17{,}169.99\, psi}$

Part 3. Determine the Column Strength of Hat-Stiffened Panel

$$\sigma_c = \sigma_{cs} - \frac{\sigma_{cs}^2}{4\pi^2 E}\left(\frac{L_e}{\rho}\right)^2 = 17{,}169.99\, psi - \frac{(17{,}169.99\, psi)^2}{4\pi^2(10.7 * 10^6\, psi)}\left(\frac{4\, in}{1.4729\, in}\right)^2 = 17{,}164.84\, psi$$

<u>**Column Strength of Hat-Stiffened Panel:**</u> $\boldsymbol{\sigma_c = 17{,}164.84\, psi}$

Part 4. Determine the Monolithic Crippling Stress

1. Hat-Panel is a multi-corner section:

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.56\left[\left(\frac{gt^2}{A}\right)\left(\frac{E}{\sigma_{cy}}\right)^{\frac{1}{2}}\right]^{0.85}$$

2. Find g_{avg}

$$g_n = \sum cuts + \sum flanges$$

$$\rightarrow g_{ST1} = (3 + 2) + (8 + 4) = 17\, , g_{ST2} = (3 + 1) + (8 + 4) = 16$$

$$g_{avg} = \frac{(1 * 17)(1 * 16)}{2} \rightarrow g_{avg} = 16.5$$

3. Find A_{tot}

$$\rightarrow A_{ST} = 0.25189\, in^2\, ; A_S = b_s * t_s = (2\, in)(0.032\, in) = 0.064\, in^2$$

$$A_{tot} = A_{ST} + A_S = 0.25189\, in^2 + 0.064\, in^2 = 0.3159\, in^2$$

4. Find $(\sigma_{cs})_M$

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.56\left[\left(\frac{gt^2}{A}\right)\left(\frac{E}{\sigma_{cy}}\right)^{\frac{1}{2}}\right]^{0.85} = 0.56\left[\left(\frac{(16.5)(0.032\, in)^2}{(0.3159\, in^2)}\right)\left(\frac{(10.7 * 10^6\, psi)}{(40{,}000\, psi)}\right)^{1/2}\right]^{0.85}$$

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.4857 < 1$$

$$(\sigma_{cs})_M = 0.4857 * \sigma_{cy} = 0.4857 * (40{,}000\ psi) = 19{,}425.9981\ psi$$

<u>**Monolithic Crippling Stress:**</u> $(\boldsymbol{\sigma_{cs}})_{\boldsymbol{M}} = \mathbf{19425.9981}\ \boldsymbol{psi}$

Part 5. Determine the Monolithic Inter-Rivet Buckling Stress

1. Find $\frac{\sigma_{ir}}{\sigma_{0.7}}$

$$(\sigma_{ir})_M = \frac{c\pi^2 E}{12(1-\vartheta^2)\sigma_{0.7}} * \left(\frac{t_s}{p}\right)^2; \quad c = 3\ for\ Brazier\ head,\ \ p = 0.75\ in$$

$$(\sigma_{ir})_M = \frac{(3)(\pi^2)(10.7 * 10^6\ psi)}{12(1-0.3^2)(39{,}000\ psi)} * \left(\frac{0.032\ in}{0.75\ in}\right)^2 = 1.2656$$

Using **Figure C7.24**: $\frac{\sigma_{ir}}{\sigma_{0.7}} = 0.93$

2. Find $(\sigma_{ir})_M$

$$\rightarrow (\sigma_{ir})_M = 0.93(\sigma_{0.7}) = 0.93(39{,}000\,psi) = 36{,}270\ psi$$

<u>**Monolithic Inter-River Buckling Stress:**</u> $(\boldsymbol{\sigma_{ir}})_{\boldsymbol{M}} = \mathbf{36{,}270}\ \boldsymbol{psi}$

Part 6. Determine the Sheet Wrinkling Stress

4. Find f

$$b_0 = \frac{b}{2} - \frac{t_w}{2} = \frac{1\ in}{2} - \frac{0.032\ in}{2} = 0.484\ in$$

$$\frac{b_0}{t_w} = \frac{0.484\ in}{0.032\ in} = 15.125; \quad \frac{p}{d} = \frac{0.75\ in}{.125\ in} = 6$$

Using **Figure 7**: $\frac{f}{t_w} = 13.3\ (extropolated\ guess)$

$$\rightarrow f = 13.3(t_w) = 13.3(0.032\ in)$$

$$\rightarrow f = 0.4256\ in$$

5. Find k_w

$$b_w = h - t_s = 2\ in - 0.032\ in = 1.968\ in$$

$$\frac{b_w}{t_w} = \frac{1.968\ in}{0.032\ in} = 61.5; \quad \frac{b_s}{t_s} = \frac{2\ in}{0.032\ in} = 62.5; \quad \frac{b_w/t_w}{b_s/t_s} = \frac{61.5}{62.5} = 0.984$$

$$\frac{f}{b_w} = \frac{0.4256\ in}{1.968\ in} = 0.2163$$

Using **Figure 6:** $k_w = 4.7$

6. Find σ_w

$$\frac{k_w \pi^2 E}{12(1-\nu^2)\sigma_{0.7}} * \left(\frac{t_s}{b_s}\right)^2 = \frac{(4.5)(\pi^2)(10.7 * 10^6 psi)}{12(1-(0.3)^2)(39{,}000\ psi)} * \left(\frac{0.032\ in}{2\ in}\right)^2 = 0.2669$$

Using **Figure C7.24:** $\frac{\sigma_w}{\sigma_{0.7}} = 0.26 < 0.9 \rightarrow OK!$

$$\rightarrow \sigma_w = 0.26(\sigma_{0.7}) = 0.26(39{,}000\ psi) = 10{,}140\ psi$$

Sheet Wrinkling Stress: $\boldsymbol{\sigma_w = 10{,}140\ psi}$

Part 7. Determine the Crippling Stress of One Hat-Stringer

1. Find $(\sigma_{cs})_{ST1}$

$$\frac{\sigma_{cs}}{\sigma_{cy}} = 0.56\left[\left(\frac{g * t^2}{A}\right)\left(\frac{E}{\sigma_{cy}}\right)^{\frac{1}{2}}\right]^{0.85} = 0.56\left[\left(\frac{(11 * 0.032\ in)^2}{(0.2518\ in^2)}\right)\left(\frac{(10.7 * 10^6 psi)}{(40{,}000\ psi)}\right)^{\frac{1}{2}}\right]^{0.85} = 0.4172 < 0.8(\sigma_{cy})$$

$$\rightarrow \frac{\sigma_{cs}}{\sigma_{cy}} = 0.4172$$

$$\sigma_{cs} = 0.4172\sigma_{cy} = 0.4172(40{,}000\ psi) = 16{,}688.3693\ psi$$

Crippling Stress of One Hat-Stringer: $\boldsymbol{(\sigma_{cs})_{ST1} = 16{,}688.3693\ psi}$

Part 8. Determine the Column Failing Stress of One-Hat Stringer

1. Find $(\sigma_c)_{ST1}$

$$L_e = \frac{L}{\sqrt{c}} = \frac{8\ in}{\sqrt{4}} = 4\ in \quad ; \quad p = 0.8035\ in$$

$$\sigma_c = \sigma_{cs} - \frac{{\sigma_{cs}}^2}{4\pi^2 E}\left(\frac{L_e}{p}\right)^2 = 16{,}688.3693\ psi - \frac{(16{,}688.3693\ psi)^2}{4\pi^2(10.7 * 10^6\ psi)}\left(\frac{4\ in}{1.4729\ in}\right)^2$$

$$\rightarrow (\sigma_c)_{ST1} = 16{,}665\ psi$$

Column Failing Stress of One Hat-Stringer: $\boldsymbol{(\sigma_{cs})_{ST1} = 16{,}665\ psi}$

Part 9. Determine the Column Failing Stress of One Hat-Stringer with Effective Width

1. Find $W_{eff}^{(0)}$

$$W_0 = 1.9\left(t\sqrt{\frac{E}{\sigma_{ST1}}}\right) = 1.9\left((0.032\ in)\sqrt{\frac{(10.7\,x10^6 psi)}{(16{,}683.1664\ psi)}}\right) = 1.48855\ in < 3in$$

$$W_1 = 0.62\left(t\sqrt{\frac{E}{\sigma_{St1}}}\right) = 0.62\left((0.032\ in)\sqrt{\frac{(10.7\ x10^6 psi)}{(16{,}683.1664\ psi)}}\right) = 0.4857 \text{ in} < 0.5 \text{ in}$$

$$W_{eff}^{(0)} = W_1 + (3)\frac{W_0}{2} = 0.4857\ in + (3)\frac{1.48855\ in}{2} = 2.97872\ in$$

2. Find $(\sigma_c)_{ST1}$

$$\left(\frac{\rho_1}{\rho_0}\right)^2 = \frac{1 + \left[1 + \left(\frac{S}{\rho_0}\right)^2\right]\frac{W_{eff}^{(0)}t}{A_{ST}}}{1 + \left(\frac{W_{eff}^{(0)}t}{A_{ST}}\right)^2} = \frac{1 + \left[1 + \left(\frac{\frac{t}{2}+\frac{h}{2}}{\rho_0}\right)^2\right]\frac{W_{eff}^{(0)}t}{A_{ST}}}{1 + \left(\frac{W_{eff}^{(0)}t}{A_{ST}}\right)^2}$$

$$\left(\frac{\rho_1}{\rho_0}\right)^2 = \frac{1 + \left[1 + \left(\frac{\frac{0.032\,in}{2}+\frac{3\,in}{2}}{(0.8035\,in)}\right)^2\right]\frac{(2.97872\,in)(0.032\,in)}{(0.2518\,in^2)}}{1 + \left(\frac{(2.97872\,in)(0.032\,in)}{(0.2518\,in^2)}\right)^2} = 1.7349$$

$$\rho_1 = \sqrt{1.7349(\rho_0)^2} = \sqrt{1.7349(0.8035\,in)^2} = 1.05839\,in$$

$$\frac{L_e}{\rho_1} = \frac{4\,in}{1.05839\,in} = 3.7793 < 20$$

$$\sigma_c = \sigma_{cs} - \frac{(\sigma_{cs})^2}{4\pi^2 E}\left(\frac{L_e}{\rho_1}\right)^2 = (16{,}688.3693\,psi) - \frac{(16{,}688.3693\,psi)^2}{(4)(\pi^2)(10.7\,x10^6 psi)}\left(\frac{4\,in}{1.05839\,in}\right)^2 = 16{,}672.44\,psi$$

<u>Column Failing Stress of Stringer 1 with Effective Width</u>: $(\boldsymbol{\sigma_c})_{ST1} = \mathbf{16{,}672.44}\ \boldsymbol{psi}$

Part 10. Determine the Column Failing Load of One Hat-Stringer with Effective Area

1. Find $A_{eff}^{(1)}$

$$A_{eff}^{(1)} = W_{eff}^{(0)} t_s = (2.97872\,in)(0.032\,in) = .09529\,in^2$$

2. Find A_{tot}

$$A_{tot} = A_{st} + A_{eff}^{(1)} = 0.2518\,in^2 + 0.09529\,in^2 = 0.34709\,in^2$$

3. Find $(P_c)_{ST1}$

$$(P_c)_{ST1} = (\sigma_c)_{ST1} * A_{tot} = (16{,}672.44\,psi)(0.34709\,in^2) = 5{,}788\,lbf$$

<u>Column Failing Stress of Stringer 1 with Effective Width</u>: $(\boldsymbol{P_c})_{ST1} = \mathbf{5{,}788}\ \boldsymbol{lbf}$

Part 11. Determine the Inter-Rivet Buckling Stress of One Hat-Stringer

1. Find $(\sigma_{ir,e})_{ST1}$

$$(\sigma_{ir,e})_{ST1} = \frac{\pi^2 E}{\left(\frac{P}{0.29t\sqrt{c}}\right)^2} = \frac{\pi^2(10.7\,x10^6 psi)}{\left(\frac{0.75\,in}{0.29(0.032\,in)\sqrt{3}}\right)^2} = 45{,}330.9\,psi$$

2. Check Elasticity with Ramberg Osgood Equation

$$\frac{E_t}{E} = \frac{1}{1+\frac{3}{7}n\left(\frac{\sigma}{\sigma_{0.7}}\right)^{n-1}} = \frac{1}{1+\frac{3}{7}(11.5)\left(\frac{45{,}330.9\,psi}{39{,}000\,psi}\right)^{11.5-1}} = 0.05682 < 1$$

$$\rightarrow inelastic;\ plasticity\ correction\ needed!$$

3. Find $(\sigma_{ir})_{ST1}$

$$\frac{p}{t} = \frac{0.75\,in}{0.032\,in} * \frac{\sqrt{4}}{\sqrt{3}} = 27.06329$$

Using **Figure C7.18**: $\boldsymbol{(\sigma_{ir})_{ST1} = 32{,}000\,psi} > (\sigma_c)_{ST1} \rightarrow OK!$

Inter-Rivet Buckling Stress of One Hat-Stringer: $\boldsymbol{(\sigma_{ir})_{ST1} = 32{,}000\ psi}$

Part 12. Determine the Panel Failing Strength and Panel Failing Load

1. Find σ_F

$$\sigma_F = \frac{\sigma_W(b_s * t_s) + min[(\sigma_{cs})_{ST1}, (\sigma_{cs})_M, \sigma_W, (\sigma_{ir})_M) * A_{ST}}{(b_s * t_s) + A_{ST}}$$

$$(\sigma_{cs})_{ST1} = 16{,}688.3693\,psi$$
$$(\sigma_{cs})_M = 19{,}425.9981\,psi$$
$$\sigma_W = 10{,}140\,psi$$
$$(\sigma_{ir})_M = 36{,}270\,psi$$

$$\sigma_W < (\sigma_{cs})_{ST1} < (\sigma_{cs})_M < (\sigma_{ir})_M$$

$$\sigma_F = \frac{(10{,}140\,psi)(2\,in * 0.032\,in) + (16{,}688.3693\,psi) * (0.025189\,in^2)}{(2\,in * 0.032\,in) + 0.025189\,in^2} = 15{,}361.6555\,psi$$

Panel Failing Strength: $\boldsymbol{\sigma_F = 15{,}361.6555\,psi}$

2. Find P_F

$$P_F = \sigma_F * A_{tot} = (15{,}361.6555\,psi)(0.31589\,in^2) = 4{,}852.5933\,lbf$$

Panel Failing Load: $\boldsymbol{P_F = 4{,}852.5933\,lbf}$

D. Theoretical Results

The theoretical analysis established the expected compressive limits for the C-stringer and Hat-stringer panel configurations by evaluating local elastic buckling behavior and cross-sectional crippling capacity. These calculations produced the baseline failure stresses and loads used to assess later numerical and experimental findings.

For the C-stringer, the predicted local elastic buckling stress was 13,455 psi, corresponding to a buckling load of 2,277 lbf. The crippling assessment produced a monolithic crippling stress of 16,969 psi and a one-stringer crippling stress of 17,168 psi. When combined, these results indicated a theoretical panel failure stress of 15,084 psi and a corresponding panel failure load of 2,623 lbf. These values established the governing theoretical limit for the C-stringer configuration.

For the Hat-stringer, the calculated local elastic buckling stress was 8,309.866 psi, with an associated buckling load of 5,788 lbf. The crippling analysis yielded a monolithic crippling stress of 19,425.9981 psi and a one-stringer crippling stress of 16,688.3693 psi. The combined theoretical evaluation produced a panel failure stress of 15,361.6555 psi and a panel failure load of 4,852.5933 lbf. These values defined the expected theoretical strength of the Hat-stringer panel.

Together, the theoretical results provided a structured baseline for both geometries. They established the minimum expected compressive capacity and served as the reference standard for validating the finite element simulations and subsequent physical testing.

~~D.~~E. Design Optimization

The theoretical phase also included the development of a MATLAB based optimization tool to automate the theoretical calculations required for identifying efficient stiffened panel configuration. The objective was to maximize failure load at varying panel dimensions. The script incorporated the governing equations of buckling, crippling, and column failure, and allowed the user to vary key geometric parameters for each panel type. The code executed several iterative evaluations to generate an optimal panel design, each with a corresponding load capacity and mass. This process enabled rapid exploration of design trade-offs in parallel with parameter combinations of optimal panel dimensions.

IV. Analytical Analysis

A. Abaqus Setup

Abaqus is a finite element analysis software used to simulate structural behavior under a wide range of loading conditions. It enables the development of detailed three-dimensional models and the application of mechanical loads to assess the resulting stresses, strains, deformations, and failure modes. Its advanced nonlinear capabilities make it suitable for aerospace applications that involve complex phenomena such as large deformations, localized plasticity, and instability.

The first step in developing an Abaqus model is the creation of the geometry. In this project, the process involved defining the cross-section of each stringer and extruding it to the full panel length. This procedure is illustrated in Figures 10 and 11.

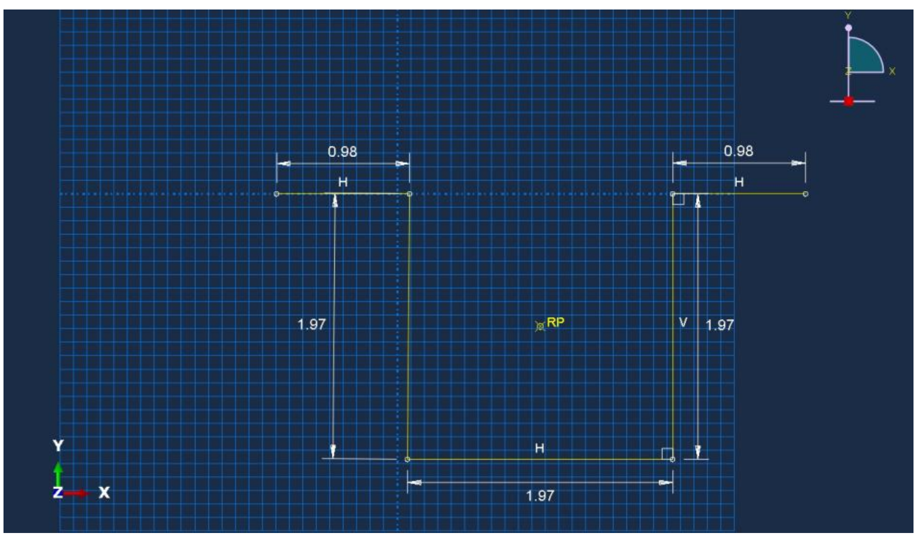

Figure 10. Panel Design Sketch

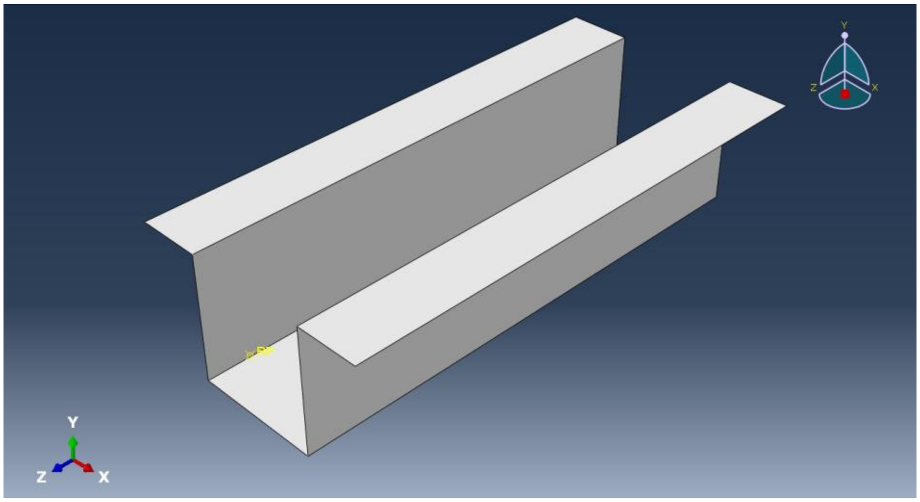

Figure 11. Extruded Sketch

Once the geometry was established, the next step was to apply the appropriate boundary conditions. Accurate representation of the experimental setup was essential to ensure meaningful simulation results. In this model, the bottom of the stringer was fully constrained, the whole top surface was permitted to deform in the z-direction. Figure 12 illustrates the boundary conditions used in the analysis.

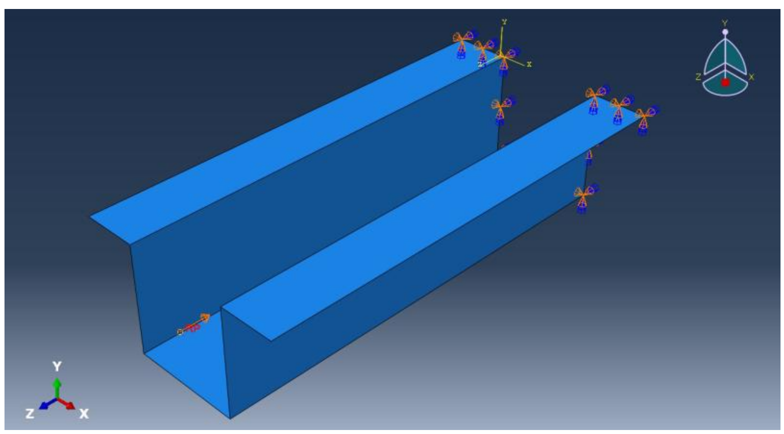

Figure 12. Boundary Conditions

Next, material properties were assigned, as they strongly influence the stress-strain response of the structure. This step included defining the material behaviors and inputting the Young's modulus and Poisson's ratio for the 2024-T3 Aluminum Alloy model. Figure 13 shows the material property window with the corresponding values.

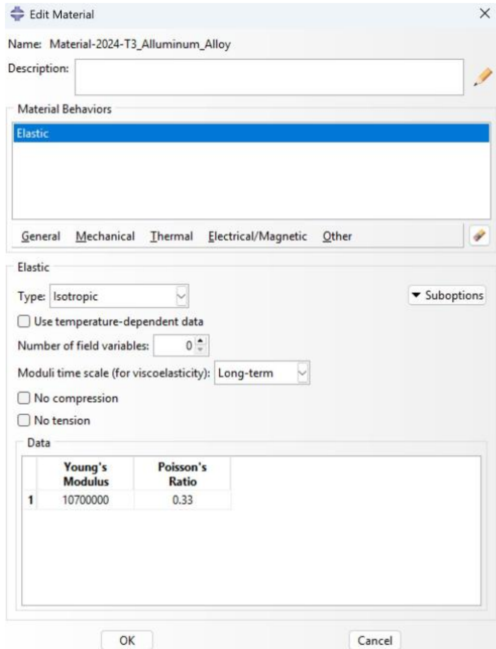

Figure 13. Material Properties

Finally, loads can be applied to the model. Figure 14 shows the applied load distribution used to replicate the experimental conditions.

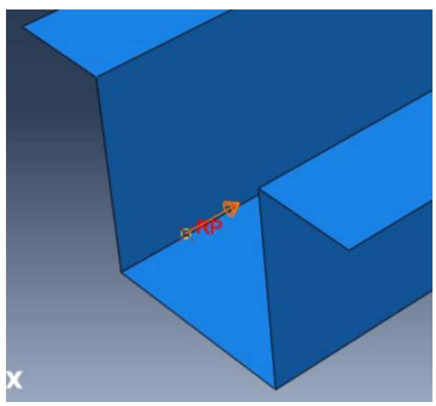

Figure 14. Z-Direction Loading

At this stage, the Abaqus model is fully defined and ready for analysis. The job can be created, submitted, and run in the solver. All resulting stress, strain, and deformation outputs are presented in the following section.

B. Abaqus Results

Abaqus simulations were performed for both the C-panel and hat-panel configurations to evaluate their structural performance under loading. The analysis provided estimates of local buckling loads, revealing the point at which each panel begins to exhibit instability in localized regions. In addition, the simulations identified the column-level failure loads, indicating the overall load-carrying capacity before global collapse. Together, these results offer a clear comparison of the structural behavior of the two panel designs and help validate the experimental findings. All results are provided in Appendix D.

V. Experimental Work & Validation

A. Experimental Setup

Before physical testing could be conducted, a critical test preparation phase was completed to ensure strict control of the boundary conditions. This process began with the fabrication of custom wooden fixtures using wood routing, during which the exact C and Hat stringer profiles were carved into wooden blocks. Potting was then performed by mixing and applying a fast-setting plaster to secure the panel ends within the routed cavities. Care was taken to ensure that each panel was positioned and formed a rigid joint with the fixture. Establishing rigid end conditions was essential to prevent premature localized failure at the panel boundaries and to ensure that structural failure would occur within the unsupported central region. After the potting material had fully cured, the panels were mounted in the Shimadzu Testing Machine (UTM) at the Advanced Composites Institute seen in Figure 15. Precise alignment during installation was a critical requirement, as the compressive load needed to be applied uniformly and symmetrically across the panel surface. This careful setup ensured that the load path remained purely axial, thereby validating the assumption of uniaxial compression used in both the theoretical calculations and numerical simulations. The resulting alignment accuracy allowed for reliable measurement of the panels' ultimate structural strength.

During testing, the panels were monitored for the key structural events outlined in Table 1. Because each panel undergoes a complete progression of deformation within a single compression test, these events were not separate experiments but observed milestones along the load–stroke curve, pictured later. Local buckling, column buckling, global buckling, and ultimate failure were identified as they occurred, allowing us to capture the corresponding loads and compare the performance of the c-stringer and hat-stringer designs. This approach ensured that the recorded behavior directly reflected the true structural response under axial compression.

Table 1. Test Matrix

Observed Event	Purpose of Test	Panel(s) Tested	What Was Measured	Resulting Failure Mode
Local Buckling	Determine the first instability in the skin or stiffener	All	First visible buckle, slight out-of-plane deformation, corresponding load	Skin buckling along web and flange
Column Buckling	Identify when the stiffener starts acting like a column and loses stability	All	Noticeable bowing of the stiffener under compression	Column-type buckling of the stringer
Global Buckling	Find the load where the entire panel begins to bend laterally	All	Overall panel curvature and sudden increase in displacement	Global panel buckling
Failure Load	Capture the maximum load the panel could withstand	All	Peak load on curve, rapid drop after failure	Full structural collapse (skin tearing or stiffener crippling)
C-Stringer Behavior	Understand behavior of C-shaped sections under compression	Ethan and Austin's	Early local buckling, progressive column buckling, global bending	Local to global buckling transition
Hat-Stringer Behavior	Understand behavior of hat-shaped sections under compression	Garv and Wesley's	Web buckling, flange rotation, higher column stability	Local buckling followed by crippling
Load-Stroke Curve	Document stiffness, buckling points, and failure	All	Load vs displacement curve showing buckling points and failure	Matches theoretical and Abaqus predictions

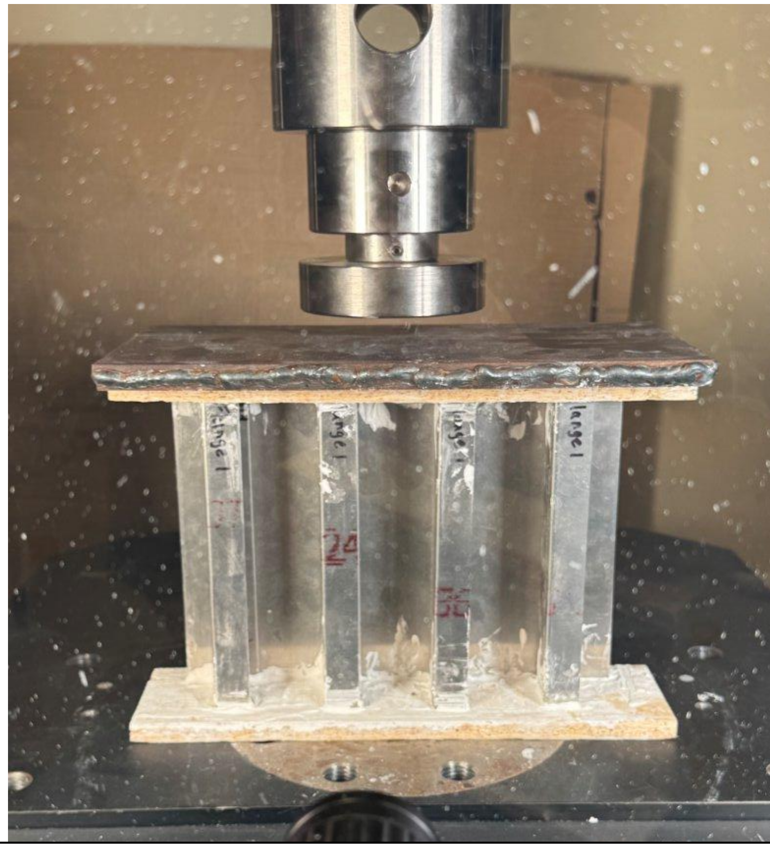

Figure x.

Figure x.

B. Experimental Results

The compression testing results were interpreted using the load displacement curves generated for all four panels. These plots identified key structural events during loading, including local buckling, column buckling, the global critical load, and the ultimate failure load. The graphical data provided a clear representation of each panel's behavior throughout the loading sequence. The individual curves for the C-stringer specimens appear in Figures 16 and 17, with the principal structural response points labeled. The results for the Hat-stringer assemblies are presented in Figures 18 and 19, which show the measured critical loads and ultimate failure points.

The two C-stringer panels reached peak failure loads of 10,365 lbf and 9,602.7 lbf. A significant issue was identified in these tests; both were conducted at a higher displacement rate than intended due to operator error. This deviation from nominal loading introduced impact-type effects and caused the panels to reach their ultimate loads more rapidly. This problem introduced substantial uncertainty into the measured results and limited the reliability of comparisons with theoretical and analytical predictions. The steep initial slope and abrupt failure visible in the C-panel curves reflect this displacement rate error. In contrast, the Hat-stringer tests were performed at the correct, slower displacement rate, resulting in representative nominal compression behavior. The first Hat panel reached a peak failure load of 9,512 lbf, and the second reached 9,256 lbf. Their close agreement indicates strong consistency and reliable data quality. The gradual force increase and extended peak region observed in the Hat-panel curves suggest a controlled and progressive failure sequence. This data set provides a dependable basis for evaluating Hat-panel performance and for validating theoretical predictions.

When the curves for all four tests are compared, as shown in the combined plot in Figure 20, all specimens confirmed that failure occurred through local buckling in the unsupported sections of the stringers. The collected experimental data also provide insight into stiffness and post-buckling response offering useful information for refining future models.

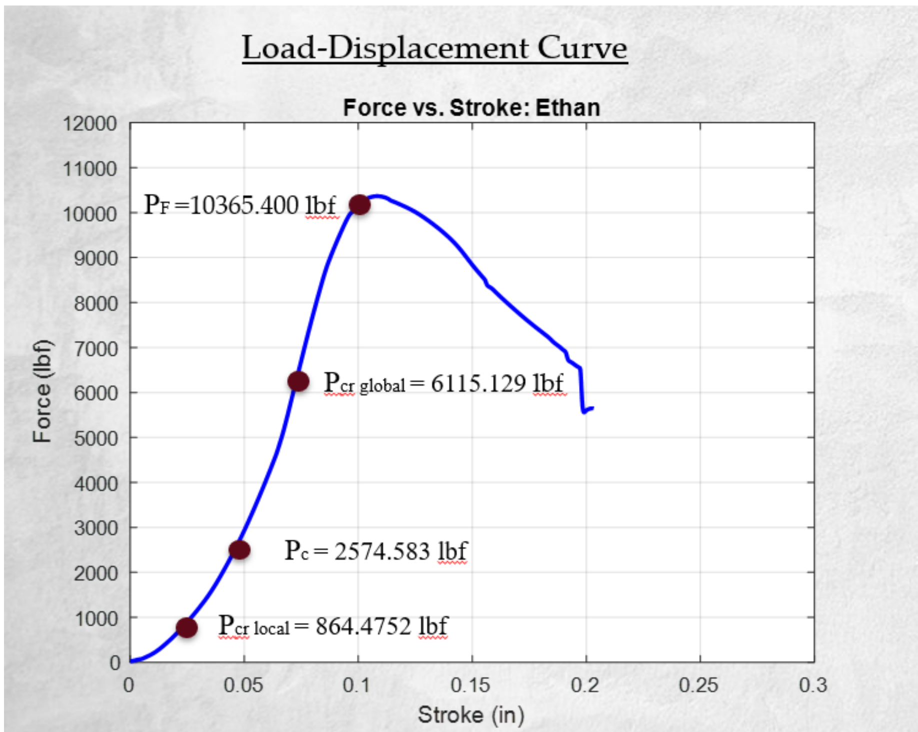

Figure 16. Ethan's C Panel Load-Displacement Curve

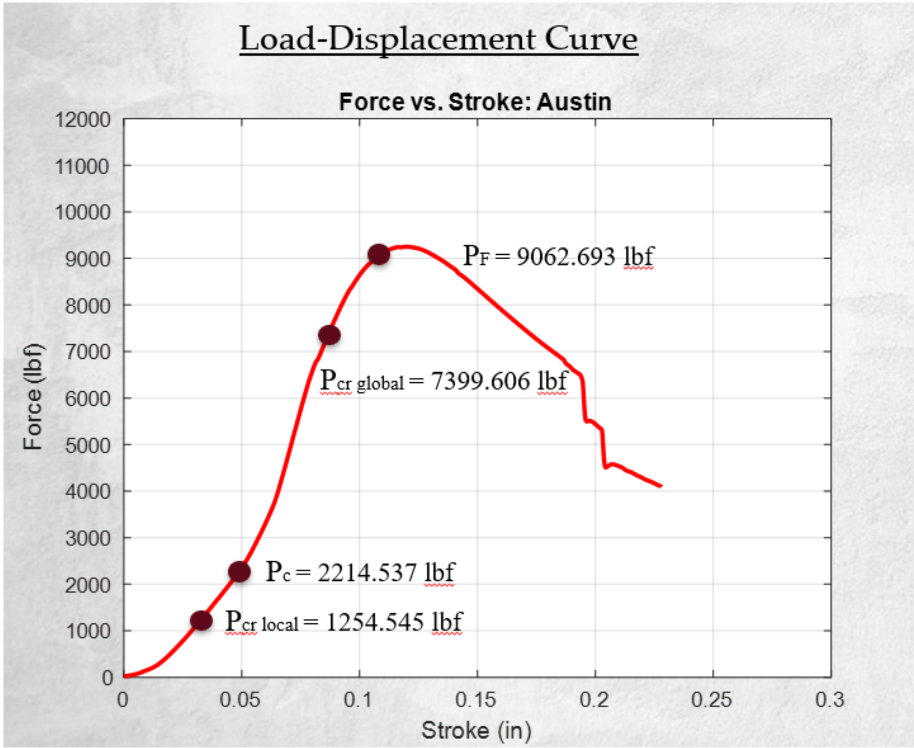

Figure 17. Austin's C Panel Load-Displacement Curve

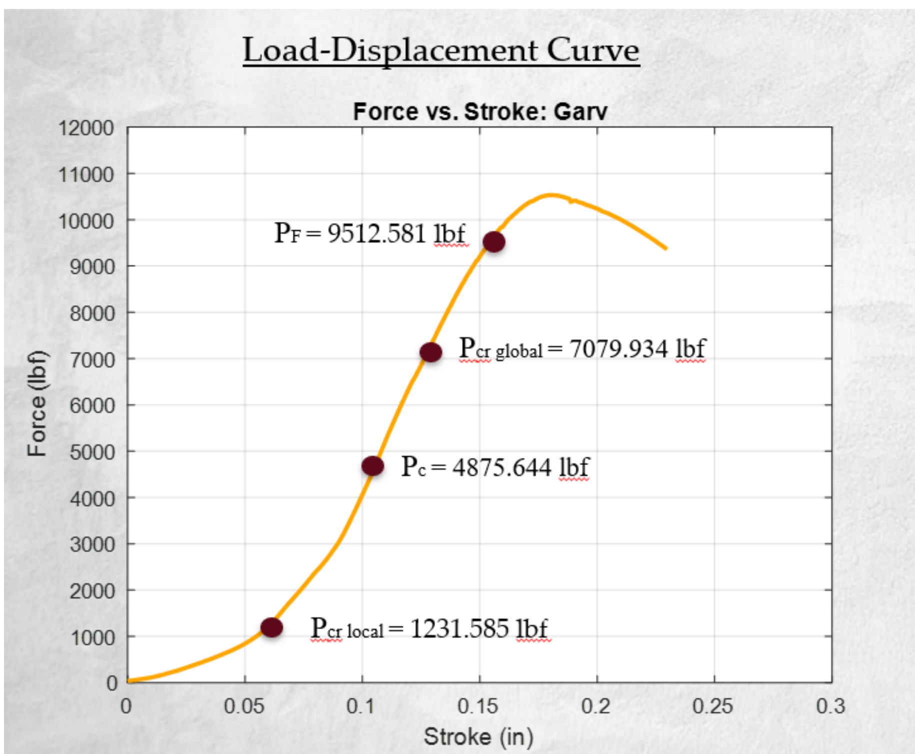

Figure 18. Garv's Hat Panel Load-Displacement Curve

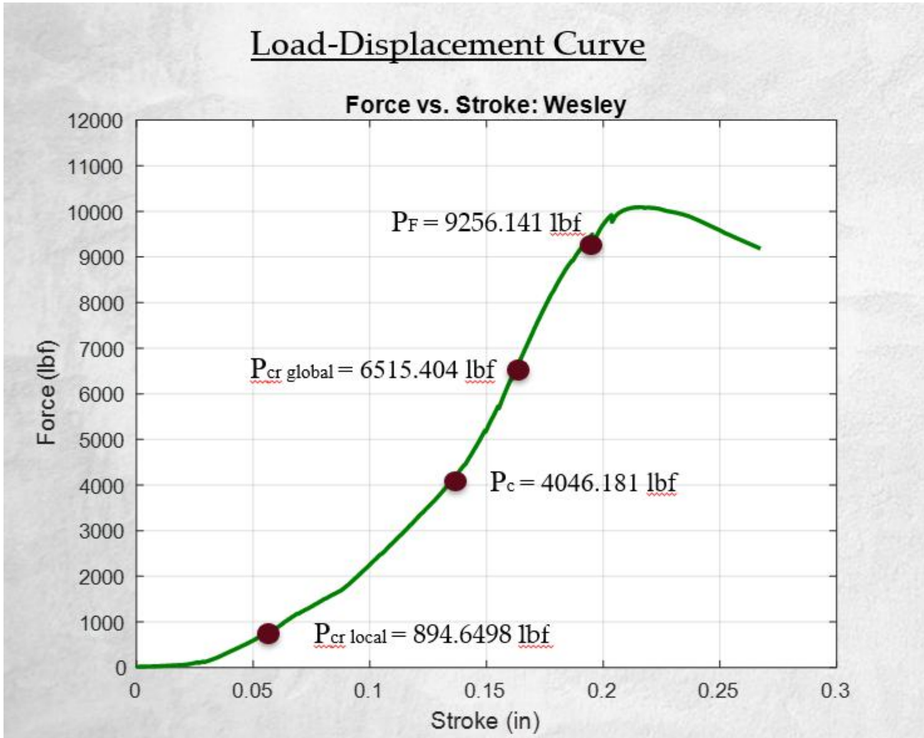

Figure 19. Wesley's Hat Panel Load-Displacement Curve

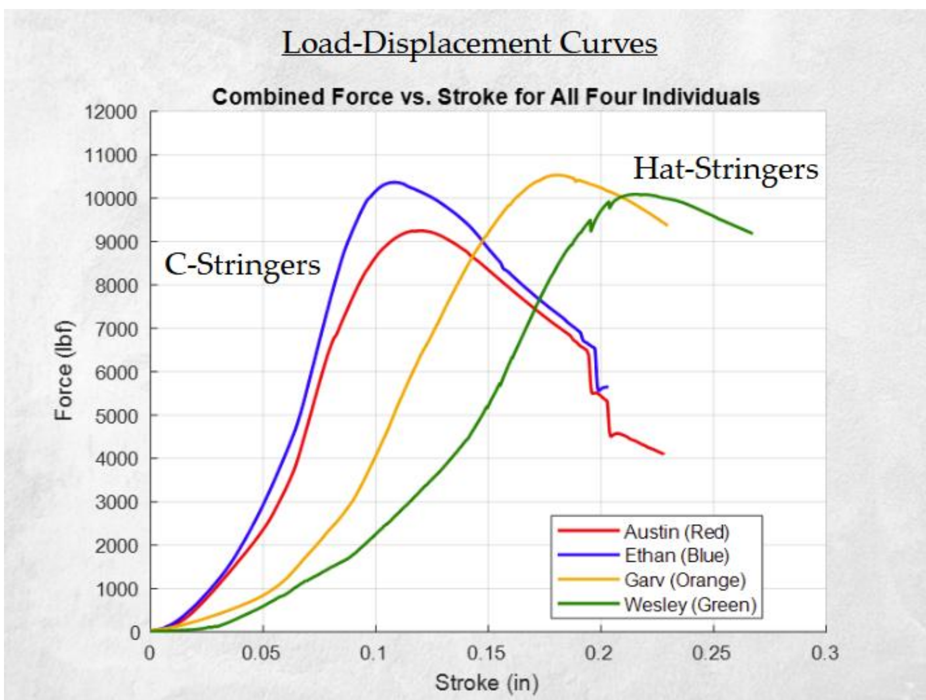

Figure 20. All Panels' Load-Displacement Curves

VI. Comparison & Validation

Validation of the structural model focused on two primary metrics: the failure mechanism and the ultimate failure load (PF). The mechanism was successfully validated across all four specimens. The observed local buckling within the stringer webs aligned with predictions from both the analytical crippling stress equations and the Finite Element Analysis (FEA) models. This agreement supports the theoretical understanding of instability behavior in the examined skin-stringer geometries. Quantitative validation of the ultimate failure load, however, produced mixed results due to inconsistencies in the testing conditions.

For the C-stringer panels, the measured peak loads (9,602 lbf to 10,365 lbf) were significantly higher than the predicted failure loads. This occurred because the panels were unintentionally subjected to a rapid loading rate, which increased the measured capacity beyond what would be expected under controlled compression. As a result, the C-panel data do not provide a reliable basis for validating the theoretical PF values. The mode of failure, characterized by local buckling, remained consistent with analytical and numerical predictions. In contrast, the Hat-stringer panels were tested under proper loading conditions and demonstrated much stronger quantitative agreement. Their average experimental peak load of 9,384 lbf served as a dependable benchmark for comparison with the analytical calculations and the FEA models. Minor discrepancies were still present and were most likely caused by manufacturing variations such as small alignment differences or uneven rivet clamping. Even with these imperfections, the close correlation observed in the Hat-stringer results confirms the accuracy of the structural model and supports the assumptions used in the pre-test analysis.

Overall, the validation efforts showed that the structural models accurately predicted the failure mechanism for both configurations. The results also highlighted the sensitivity of quantitative comparisons to loading-rate control and consistency in end conditions. The Hat-stringer results, which were collected under appropriate testing conditions, provided the strongest validation and confirmed that the project’s structural performance objectives were achieved.

In addition to confirming the failure modes, the load-level comparisons across all panels provided further insight into how each configuration behaved under compression. As summarized in Table 2, each panel displayed distinct structural events during testing, including the onset of local buckling, the development of column buckling, the emergence of global instability, and the final peak failure load. These events followed the sequence predicted by both the analytical model and the FEA simulations. The Hat-stringer panels showed higher column buckling loads, indicating greater stiffness during the intermediate stages of deformation. The C-stringer panels, however, demonstrated higher overall strength. They reached the largest peak loads and ultimately proved to be the stronger configuration, despite showing earlier local buckling due to their geometry.

The measured values also reflected the general trends predicted by the analytical and numerical tools. The Hat-stringer panels reached total failing loads between 9,256 lbf and 9,512 lbf, while the C-stringer panels achieved the highest failure loads within the range of 9,600 to 10,400 lbf. These results support the expected behavior, even though the rapid loading applied to the C panels increased their recorded capacities.

Table 2. Test Results Summary

Column Failing Loads

Panel	Homework Values (Theoretical)	MATLAB Work (Optimization)	Test Results (Experimental)	Abaqus (Analytical)
Garv's Hat-Stringer	5788 lbf	5150.50 lbf	4875.64 lbf	5911.91 lbf
Ethan's C-Stringer	2277 lbf	2450.48 lbf	2574.58 lbf	2893.00 lbf
Wesley's Hat-Stringer	5788 lbf	5150.50 lbf	4046.18 lbf	5911.91 lbf
Austin's C-Stringer	2277 lbf	3950.48 lbf	2214.54 lbf	2893.00 lbf

Total Failing Loads

Panel	Homework Values (Theoretical)	MATLAB Work (Optimization)	Test Results (Experimental)	Abaqus (Analytical)
Garv's Hat-Stringer	9705.18 lbf	10301.01 lbf	9512.58 lbf	11823.82 lbf
Ethan's C-Stringer	10492.00 lbf	10521.92 lbf	10365.40 lbf	11572.00 lbf
Wesley's Hat-Stringer	9705.18 lbf	10301.01 lbf	9256.14 lbf	11823.82 lbf
Austin's C-Stringer	10492.00 lbf	10521.92 lbf	9602.69 lbf	11572.00 lbf

Local Buckling Loads

Panel	Homework Values (Theoretical)	MATLAB Work (Optimization)	Test Results (Experimental)	Abaqus (Analytical)
Garv's Hat-Stringer	2093.17 lbf	N/A	1231.59 lbf	2170.40 lbf
Ethan's C-Stringer	1264.12 lbf	N/A	864.48 lbf	672.69 lbf
Wesley's Hat-Stringer	2093.17 lbf	N/A	894.65 lbf	2170.40 lbf
Austin's C-Stringer	1264.12 lbf	N/A	1254.55 lbf	672.69 lbf

VII. Summary and Conclusions

The objectives of this structural design project were met by advancing two stiffener geometries, the C-stringer and Hat-stringer, through a complete sequence of design, fabrication, testing, and analytical evaluation. Four assemblies were produced, and the resulting manufacturing data demonstrated a steep learning curve, with rapid improvements in efficiency for complex operations such as metal forming and riveting. Experimental testing confirmed a central theoretical expectation, specifically structural failure consistently initiated in the same manner for both stiffener configurations. However, a significant factor emerged during data evaluation. The Hat-stringer panels, tested at the specified low loading rate, generated consistent peak loads averaging approximately 9384 lbf. In contrast, the C-stringer panels were tested at an unintended high loading rate, producing elevated peak loads of up to 10,365 lbf. This discrepancy indicates that the apparent strength advantage in the C-stringer data is not valid for comparison against static theoretical predictions. This distinction highlights the necessity for strict control of test conditions to ensure reliable results and illustrates the practical challenges associated with aligning analytical models with physical behavior when unexpected variables occur. Ultimately, the project delivered substantial analytical and experimental insight into aerospace structural evaluation, confirming the expected response mode while underscoring the inherent difficulty of achieving perfect agreement between theoretical and measured performance.

Appendix A. Design Details

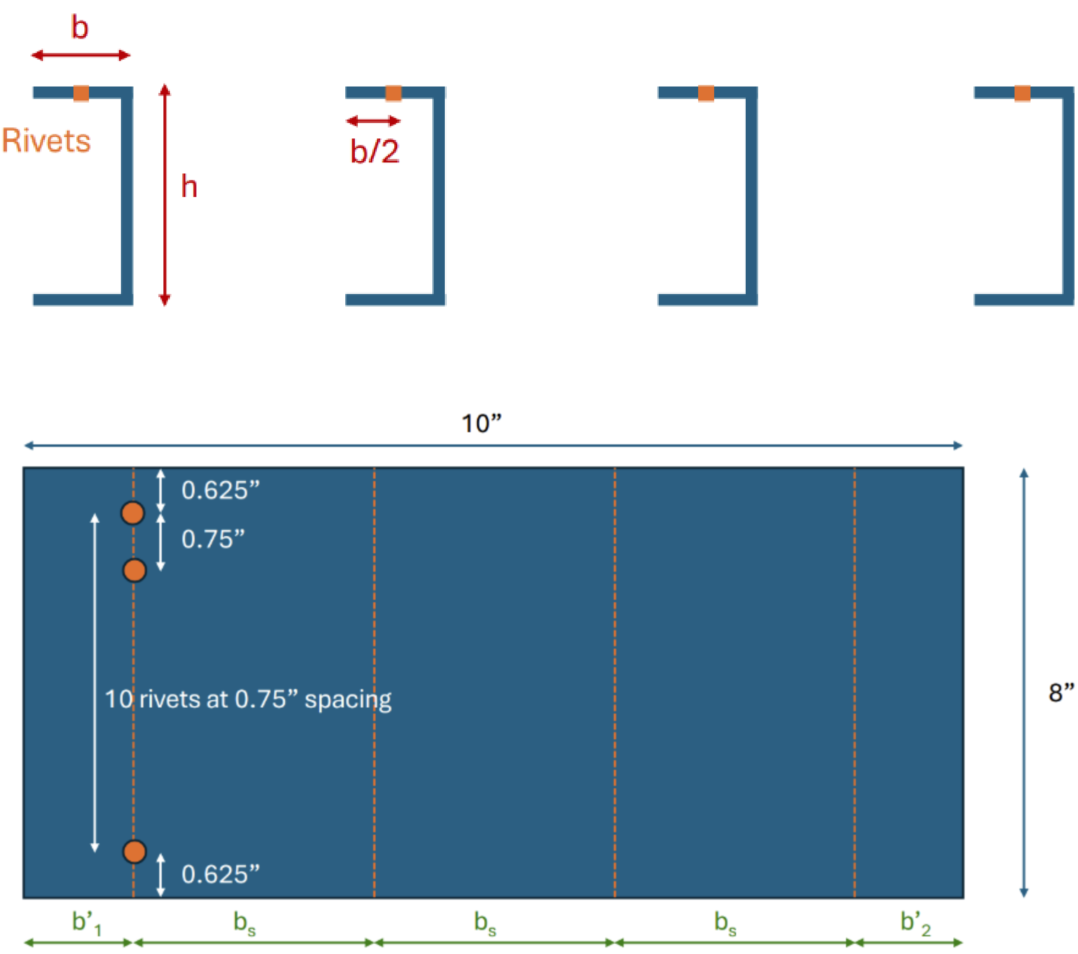

Figure 21. Complete C-Panel Given Design

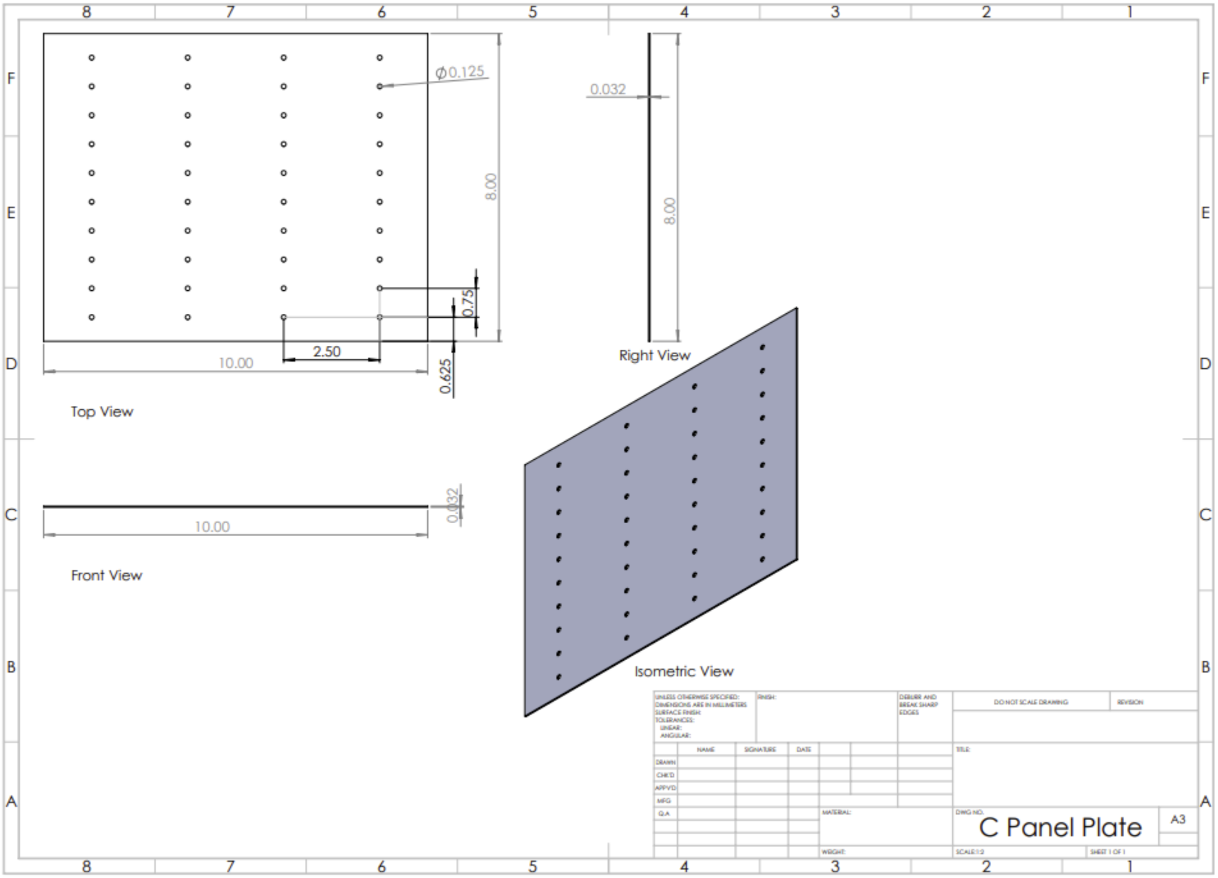

Figure 22. C-Panel Engineering Drawing

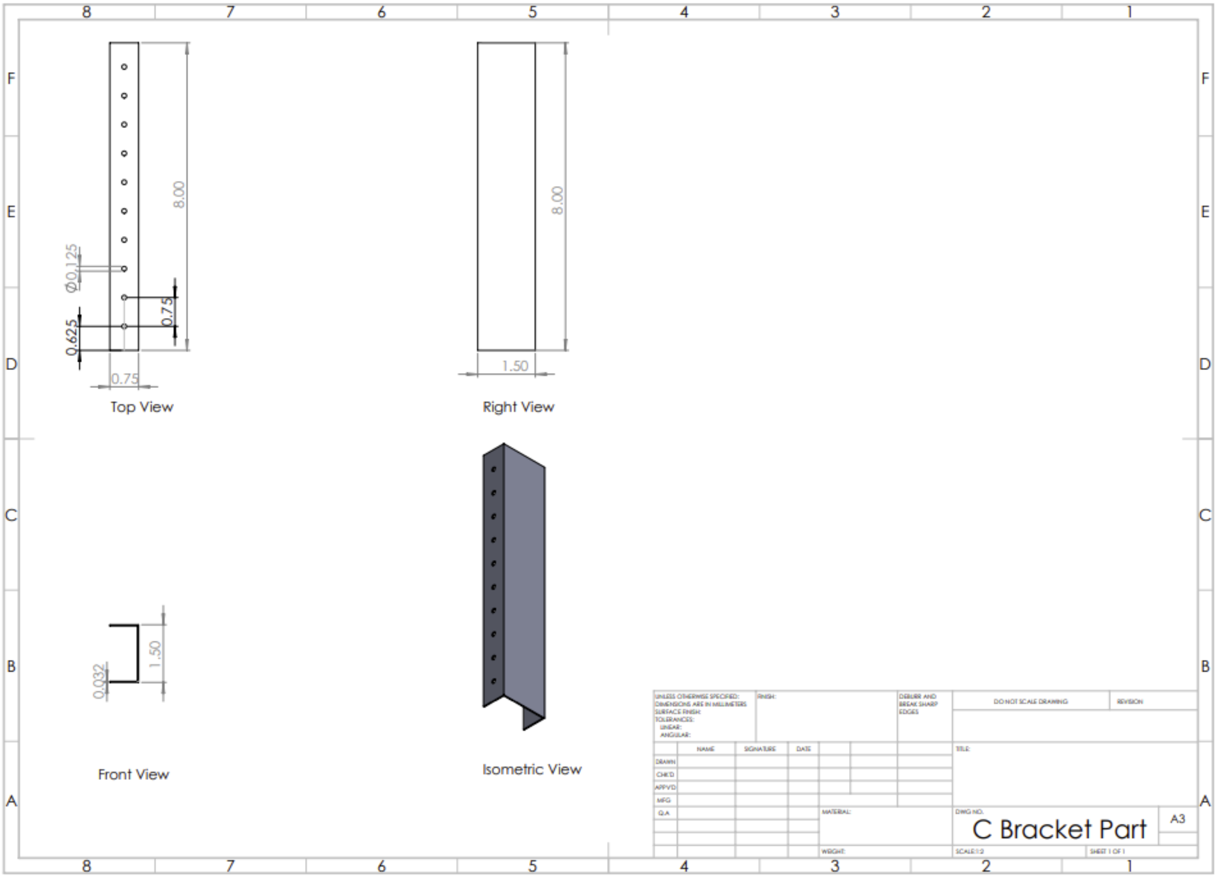

Figure 23. C-Stringer Engineering Drawing

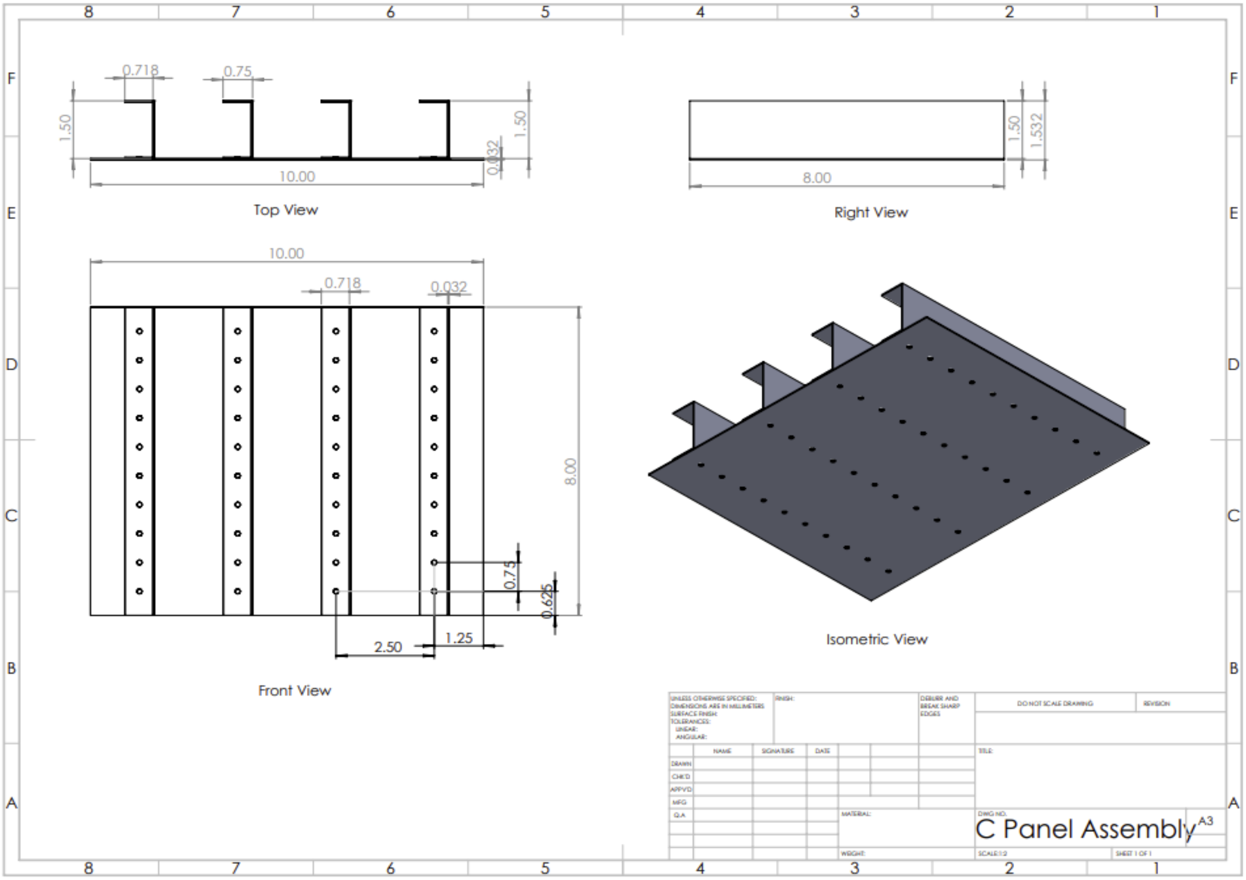

Figure 24. Assembled C-Panel Engineering Drawing

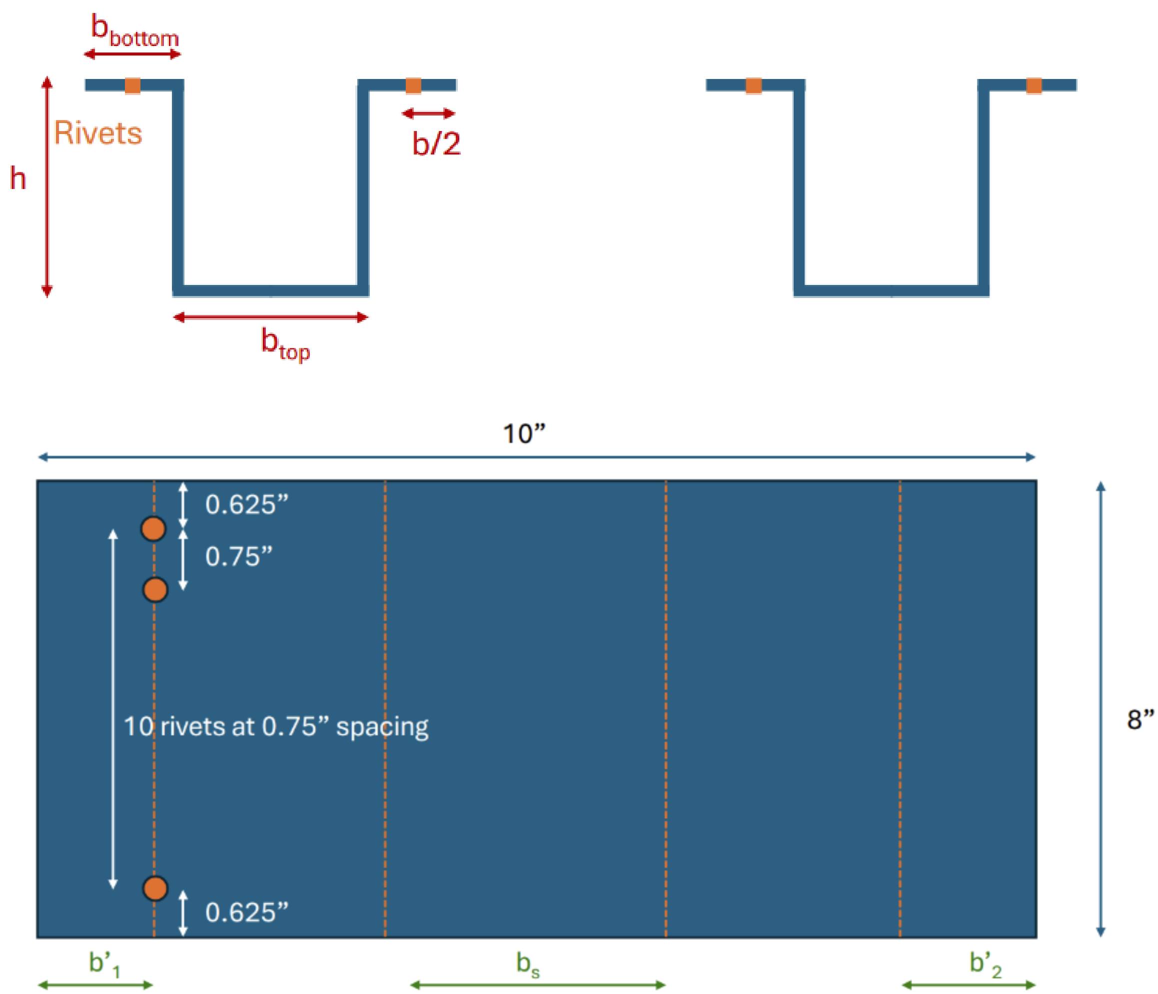

Figure 25. Complete Hat-Panel Given Design

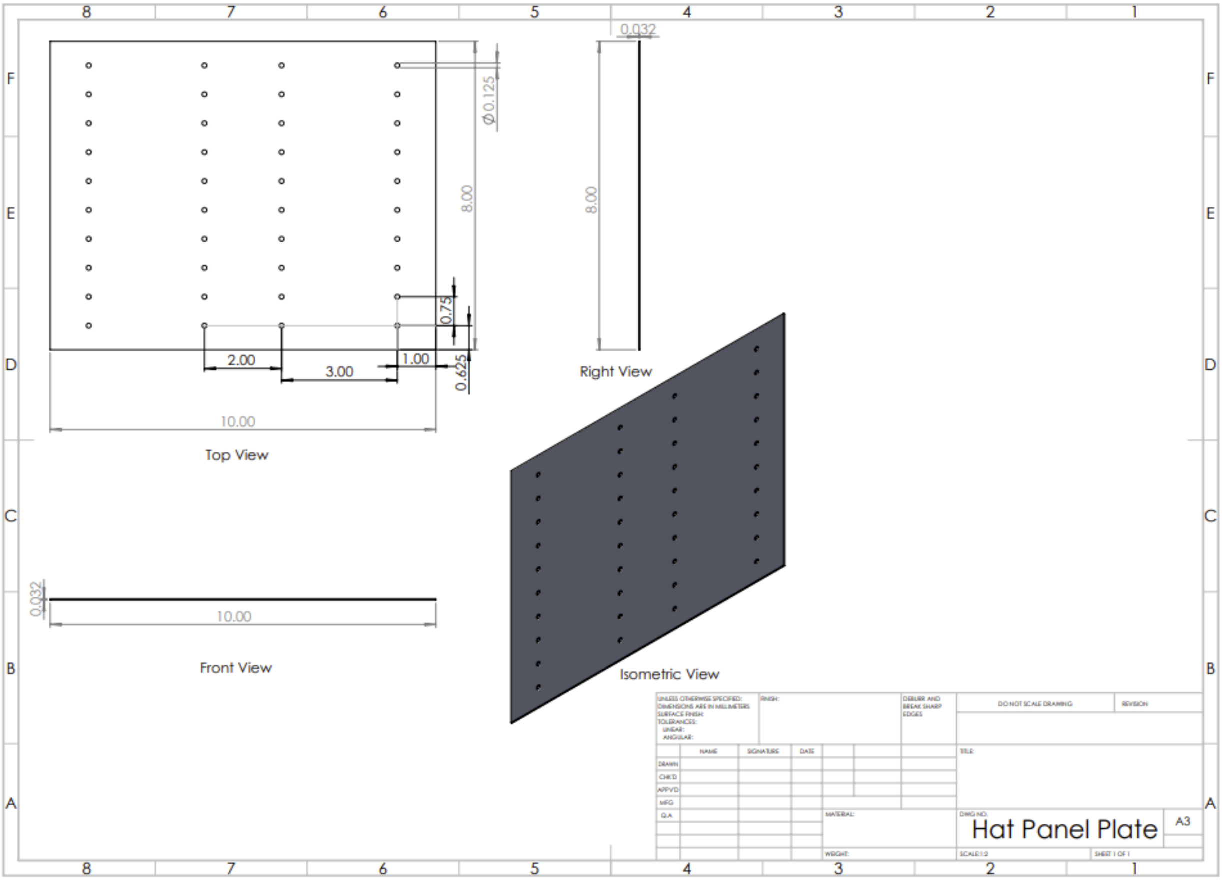

Figure 26. Hat-Panel Engineering Drawing

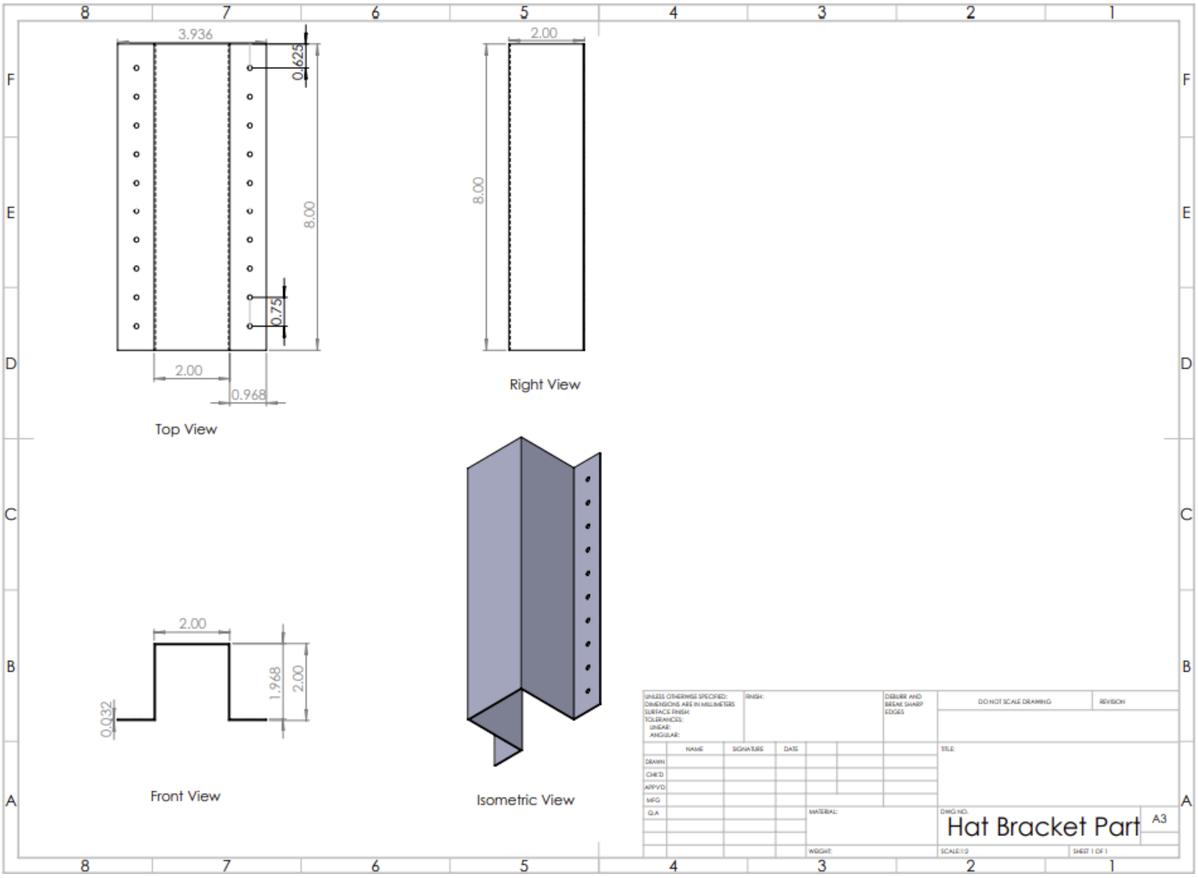

Figure 27. Hat-Stringer Engineering Drawing

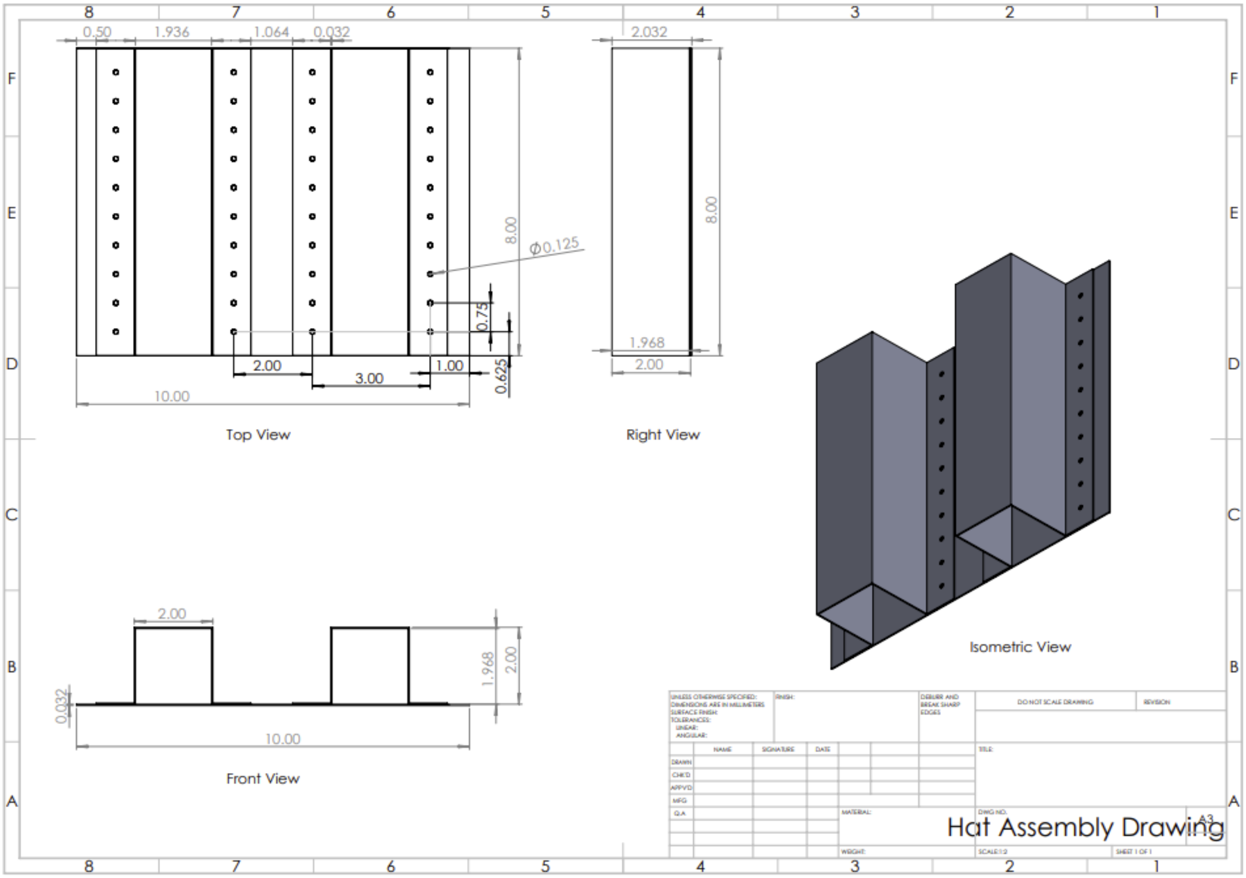

Figure 28. Assembled Hat-Panel Engineering Drawing

Appendix B. Codes for Analysis

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```
clc;
clear;
close all;

%% Notes %%

% The boundary conditions are assumed to be clamped.
% This code was written for a panel with 4 C-Stringers

%% Inputs %%

% Material Properties - Aluminum Alloy 2024-T3

material.E = 10.7e6;                % Young's Modulus, Unit: psi
material.nu = 0.3;                  % Poisson's Ratio
material.scy = 40e3;                % Yielding Stress, Unit: psi
material.s0p7 = 39e3;               % Sigma_0.7, Unit: psi
material.n = 11.5;                  % Material Property
material.wpv = 0.26;                % Weight Per Volume, Unit: lb/in3

% C-Stringer Dimensions

ST.h = 1.5;                          % Stringer Height, Unit: in
ST.b = 0.75;                         % Stringer Width, Unit: in
ST.t = 0.032;                        % Stringer Thickness, Unit: in


% Skin Panel Dimensions

panel.L = 8;                       % Skin Panel Length, Unit: in
panel.W = 10;                      % Skin Panel Width, Unit: in
panel.t = 0.032;                   % Skin Panel Thickness, Unit: in

% Assembly Details

assembly.bp1 = 1.25;                  % Left free-edge distance, Unit: in
assembly.bs = 2.5;                    % Stringer Spacing, Unit: in
assembly.bp2 = 1.25;                  % Right free-edge distance: Unit: in
assembly.nST = 4;                     % Number of Stringers
assembly.c = 4;                       % Boundary Condition
assembly.Le = panel.L ./ sqrt(assembly.c);

% Optimization Constraints and Variables (I have no idea what these are?)

opt.h = 0.2;                       % Stringer Height, Unit: in
opt.b = 0.2;                       % Stringer Width, Unit: in

% Optimization Domains

domain.h = linspace(ST.h - opt.h, ST.h + opt.h, 100);
domain.b = linspace(ST.b - opt.b, ST.b + opt.b, 100);
```

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```
%% Optimization Loop %%
for i = 1:size(domain.h,2)

    for j=1:size(domain.b,2)

        % Find the Geometric Properties ( Area, Centroid Loca., MOI)

        % Area

        A_Flange_ST(i,j) = domain.b(j) .* ST.t;
        A_Web_ST(i,j) = ( domain.h(i) - (2.* ST.t) ) .* ST.t;
        A_ST(i,j) = ( 2 .* A_Flange_ST(i,j) ) + A_Web_ST(i,j);
        Ay(i,j) = ( A_Flange_ST(i,j) .* ( ST.t ./ 2) ) + ( A_Web_ST(i,j) .* ( domain.h(i) ↙
./ 2)) ...
            + (A_Flange_ST(i,j) .* ( domain.h(i) - (ST.t ./2) ));

        % Centroid

        yc(i,j) = Ay(i,j) ./ A_ST(i,j);

         % Moment of Inertia

         IX_Flange(i,j) = (1/12) .* (domain.b(j)) .* ((ST.t).^3);
         IX_Web(i,j) = (1/12) .* (ST.t) .* ( (domain.h(i) - (2 .* ST.t) ) .^3 );

         I(i,j) = (IX_Flange(i,j) + (A_Flange_ST(i,j) .* (yc(i,j) - (ST.t ./2)).^2 )) ...
             + (IX_Web(i,j) + (A_Web_ST(i,j) .* (yc(i,j) - (domain.h(i) ./2)).^2)) ...
             + (IX_Flange(i,j) + (A_Flange_ST(i,j) .* (yc(i,j) - (domain.h(i) - (ST.t . ↙
/2)).^2)));

         % Find the Crippling Stress

         g = 5;

         scs_scy(i,j) = 3.2 .* ( ((g .* (ST.t).^2) ./ A_ST(i,j)) .* ( (material.E ./ ↙
material.scy) .^ (1/3))) .^ (0.75);

         if scs_scy(i,j) > 0.8
            scs_scy(i,j) = 0.8;
        end
        scs(i,j) = scs_scy(i,j) .* material.scy;

        rho_0 (i,j) = sqrt( I(i,j) ./ A_ST(i,j));
        slender(i,j) = assembly.Le ./ rho_0(i,j);

        if slender(i,j) < 20
            sc_0(i,j) = scs(i,j);
        else
            sc_0(i,j) = (scs(i,j) - (((scs(i,j)).^2)/(4 .* pi^2 .* material.E)) .* ↙
((slender(i,j))^2));
```

```
        end

        % Find the Effective Area

        Aeff(i,j) = 0;

        w1(i,j) = 0.62 .* (panel.t .* sqrt(material.E ./ sc_0(i,j)));
        w(i,j) = 1.9 .* (panel.t .* sqrt(material.E ./ sc_0(i,j)));

        for k = 1:size(assembly.nST,2)
            if k == 1
                if w1(i,j) > assembly.bp1
                    weff = assembly.bp1 + 0.5 .* w(i,j);
                else
                    weff = w(i,j);
                end
                Aeff(i,j) = Aeff(i,j) + weff .* panel.t;

            elseif k == size(assembly.nST,2)
                if w1(i,j) > assembly.bp2
                    weff = assembly.bp2 + 0.5 .* w(i,j);
                else
                    weff = w(i,j);
                end
                Aeff(i,j) = Aeff(i,j) + weff .* panel.t;
            else
                weff = w(i,j);
                Aeff(i,j) = Aeff(i,j) + weff .* panel.t;
            end
        end
        % Find the Compressive Load for Column Failure

        Aeff_tot(i,j) = (A_ST(i,j) .* assembly.nST) + Aeff(i,j);
        Pc(i,j) = sc_0(i,j) .* Aeff_tot(i,j);

        % Find the Column Failure Load per Panel Weight

        A_tot(i,j) = (A_ST(i,j) .* assembly.nST) + (panel.W .* panel.t);
        Pc_weight(i,j) = Pc(i,j) ./ (A_tot(i,j) .* panel.L .* material.wpv);

    end
end

%% Find the Optimized Design Parameters %%

[B,H] = meshgrid(domain.b,domain.h);

%For the Maximum Load Only

[M1,I1] = max(Pc);
```

```
[M2,I2] = max(M1);

fprintf('=== For the maximum load only ===\n')
Pc_max = Pc(I1(I2),I2);
fprintf('Pc_max=%.2f lb\n',Pc_max)

h_max = H(I1(I2),I2);
fprintf('h at Pc_max=%.2f in\n',h_max)

b_max = B(I1(I2),I2);
fprintf('b at Pc_max=%.2f in\n',b_max)

% For the maximum load per panel weight

[M1_weight,I1_weight]=max(Pc_weight);

[M2_weight,I2_weight]=max(M1_weight);

fprintf('=== For the maximum load per panel weight ===\n')
Pc_weight_max = Pc_weight(I1(I2_weight),I2_weight);
fprintf('Pc per weight_max=%.2f lb/lb \n',Pc_weight_max)

h_max_weight = H(I1_weight(I2_weight),I2_weight);
fprintf('h at Pc per weight_max=%.2f in\n',h_max_weight)

b_max_weight = B(I1_weight(I2_weight),I2_weight);
fprintf('b at Pc per weight_max=%.2f in\n',b_max_weight)

%%% Plot the result %%%

figure
surfc(B,H,Pc)
colorbar
xlabel('b (in)', 'FontName', 'Times','FontSize',15)
ylabel('h (in)', 'FontName', 'Times','FontSize',15)
zlabel('Pc (lb)', 'FontName', 'Times','FontSize',15)
title('Column Failure Load', 'FontName', 'Times','FontSize',15)

figure
surfc(B,H,Pc_weight)
colorbar
xlabel('b (in)', 'FontName', 'Times','FontSize',15)
ylabel('h (in)', 'FontName', 'Times','FontSize',15)
zlabel('Pc per weight (lb/lb)', 'FontName', 'Times','FontSize',15)
title('Column Failure Load per Panel Weight', 'FontName','Times','FontSize',15)
```

Figure ~~x~~31. C-Panel Optimization Code

```
clc;
clear;
close all;

%% Notes %%

% The boundary conditions are assumed to be clamped.
% This code was written for a panel with 2 Hat-Stringers

%% Inputs %%

% Material Properties - Aluminum Alloy 2024-T3

material.E = 10.7e6;              % Young's Modulus, Unit: psi
material.nu = 0.3;                % Poisson's Ratio
material.scy = 40e3;              % Yielding Stress, Unit: psi
material.s0p7 = 39e3;             % Sigma_0.7, Unit: psi
material.n = 11.5;                % Material Property
material.wpv = 0.26;               % Weight Per Volume, Unit: lb/in3

% Hat-Stringer Dimensions

ST.h = 2;                         % Stringer Height, Unit: in
ST.b = 4;                         % Stringer Width, Unit: in
ST.t = 0.032;                     % Stringer Thickness, Unit: in


% Skin Panel Dimensions

panel.L = 10;                      % Skin Panel Length, Unit: in
panel.W = 8;                     % Skin Panel Width, Unit: in
panel.t = 0.032;                  % Skin Panel Thickness, Unit: in

% Assembly Details

assembly.bp1 = 1;                 % Left free-edge distance, Unit: in
assembly.bs = 2;                  % Stringer Spacing, Unit: in
assembly.bp2 = 1;                 % Right free-edge distance: Unit: in
assembly.nST = 2;                 % Number of Stringers
assembly.c = 4;                   % Boundary Condition
assembly.Le = panel.L ./ sqrt(assembly.c);

% Optimization Constraints and Variables (I have no idea what these are?)

opt.h = 0.2;                      % Stringer Height, Unit: in
opt.b = 0.2;                      % Stringer Width, Unit: in

% Optimization Domains

domain.h = linspace(ST.h - opt.h, ST.h + opt.h, 100);
domain.b = linspace(ST.b - opt.b, ST.b + opt.b, 100);
```

```
%% Optimization Loop %%

for i = 1:size(domain.h,2)

    for j=1:size(domain.b,2)

        % Find the Geometric Properties (Area, Centroid Locat., MOI)

        % Area, Units: in2

        A_FLANGE_ST(i,j) = (domain.b(j) - ST.t) .* ST.t;
        A_WEB_ST(i,j) = domain.h(i) .* ST.t;
        A_TOP_ST(i,j) = (domain.b(j) - 2.*ST.t) .* ST.t;
        A_ST(i,j) = (2 .* A_FLANGE_ST(i,j)) + (2 .* A_WEB_ST(i,j)) + A_TOP_ST(i,j);
        Ay(i,j) = (2 .* ((A_FLANGE_ST(i,j).* 0.5 .* ST.t)) + (2 .* ((A_WEB_ST(i,j).* 0.5 ↙
.* domain.h(i) )))) + (A_TOP_ST(i,j) .* (domain.h(i) - (0.5 .* ST.t)));

        % Centroid

        yc(i,j) = Ay(i,j) ./ A_ST(i,j);

        % Moment of Inertia

        IX_FLANGE(i,j) = (1/12) .* (domain.b(j) - ST.t) .* (ST.t).^3;
        IX_WEB(i,j) = (1/12) .* (ST.t) .* (domain.h(i)).^3;
        IX_TOP(i,j) = (1/12) .* (domain.b(j) - (2 .* ST.t)) .* (ST.t).^3;

        I(i,j) = ( 2 .* (IX_FLANGE(i,j) + A_FLANGE_ST(i,j) .* ( (yc(i,j) - (0.5 .* ST.t). ↙
^2) ) )) + ( 2.* (IX_WEB(i,j) + ...
                    A_WEB_ST(i,j) .* ( (yc(i,j) - (0.5 .* domain.h(i) ).^2) ) )) +  ( ↙
IX_TOP(i,j) + A_TOP_ST(i,j) .* ...
                    ((yc(i,j)) - (domain.h(i) - (0.5 .* ST.t)).^2));


        % Find the Crippling Stress

        g = 11;

        scs_scy(i,j) = 0.56 .* ( ((g .* (ST.t).^2) ./ A_ST(i,j)) .* ( (material.E ./ ↙
material.scy) .^ (1/2))) .^ (0.85);

        if scs_scy(i,j) > 0.8
            scs_scy(i,j) = 0.8;
        end
        scs(i,j) = scs_scy(i,j) .* material.scy;

        rho_0 (i,j) = sqrt( I(i,j) ./ A_ST(i,j));
        slender(i,j) = assembly.Le ./ rho_0(i,j);

        if slender(i,j) < 20
```

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```
            sc_0(i,j) = scs(i,j);
        else
            sc_0(i,j) = (scs(i,j) - (((scs(i,j)).^2)/(4 .* pi^2 .* material.E)) .*
((slender(i,j))^2));
        end

        % Find the Effective Area

        Aeff(i,j) = 0;

        w1(i,j) = 0.62 .* (panel.t .* sqrt(material.E ./ sc_0(i,j)));
        w(i,j) = 1.9 .* (panel.t .* sqrt(material.E ./ sc_0(i,j)));

        for k = 1:size(assembly.nST,2)
            if k == 1
                if w1(i,j) > assembly.bp1
                    weff = assembly.bp1 + 0.5 .* w(i,j);
                else
                    weff = w(i,j);
                end
                Aeff(i,j) = Aeff(i,j) + weff .* panel.t;

            elseif k == size(assembly.nST,2)
                if w1(i,j) > assembly.bp2
                    weff = assembly.bp2 + 0.5 .* w(i,j);
                else
                    weff = w(i,j);
                end
                Aeff(i,j) = Aeff(i,j) + weff .* panel.t;
            else
                weff = w(i,j);
                Aeff(i,j) = Aeff(i,j) + weff .* panel.t;
            end
        end

        % Find the Compressive Load for Column Failure

        Aeff_tot(i,j) = (A_ST(i,j) .* assembly.nST) + Aeff(i,j);
        Pc(i,j) = sc_0(i,j) .* Aeff_tot(i,j);

        % Find the Column Failure Load per Panel Weight

        A_tot(i,j) = (A_ST(i,j) .* assembly.nST) + (panel.W .* panel.t);
        Pc_weight(i,j) = Pc(i,j) ./ (A_tot(i,j) .* panel.L .* material.wpv);

    end
end

%% Find the Optimized Design Parameters %%

[B,H] = meshgrid(domain.b,domain.h);
```

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```
%For the Maximum Load Only

[M1,I1] = max(Pc);

[M2,I2] = max(M1);

fprintf('=== For the maximum load only ===\n')
Pc_max = Pc(I1(I2),I2);
fprintf('Pc_max=%.2f lb\n',Pc_max)

h_max = H(I1(I2),I2);
fprintf('h at Pc_max=%.2f in\n',h_max)

b_max = B(I1(I2),I2);
fprintf('b at Pc_max=%.2f in\n',b_max)

% For the maximum load per panel weight

[M1_weight,I1_weight]=max(Pc_weight);

[M2_weight,I2_weight]=max(M1_weight);

fprintf('=== For the maximum load per panel weight ===\n')
Pc_weight_max = Pc_weight(I1(I2_weight),I2_weight);
fprintf('Pc per weight_max=%.2f lb/lb \n',Pc_weight_max)

h_max_weight = H(I1_weight(I2_weight),I2_weight);
fprintf('h at Pc per weight_max=%.2f in\n',h_max_weight)

b_max_weight = B(I1_weight(I2_weight),I2_weight);
fprintf('b at Pc per weight_max=%.2f in\n',b_max_weight)

%%% Plot the result %%%

figure
surfc(B,H,Pc)
colorbar
xlabel('b (in)', 'FontName', 'Times','FontSize',15)
ylabel('h (in)', 'FontName', 'Times','FontSize',15)
zlabel('Pc (lb)', 'FontName', 'Times','FontSize',15)
title('Column Failure Load', 'FontName', 'Times','FontSize',15)

figure
surfc(B,H,Pc_weight)
colorbar
xlabel('b (in)', 'FontName', 'Times','FontSize',15)
ylabel('h (in)', 'FontName', 'Times','FontSize',15)
zlabel('Pc per weight (lb/lb)', 'FontName', 'Times','FontSize',15)
title('Column Failure Load per Panel Weight', 'FontName','Times','FontSize',15)
```

Figure ~~x~~32. Hat-Panel Optimization Code

Appendix C. Analysis Results Generated by Codes

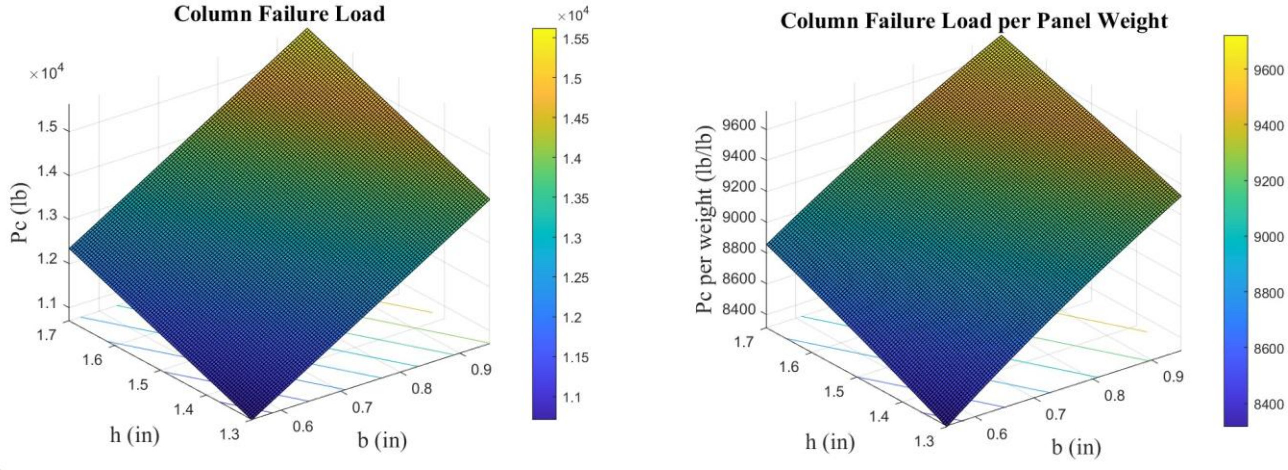

Figure 35. C-Panel Optimization Graph

```
=== For the maximum load only ===
Pc_max=15621.92 lb
h at Pc_max=1.70 in
b at Pc_max=0.95 in
=== For the maximum load per panel weight ===
Pc per weight_max=9721.02 lb/lb
h at Pc per weight_max=1.70 in
b at Pc per weight_max=0.95 in
```

Figure 36. C-Panel Optimization Results

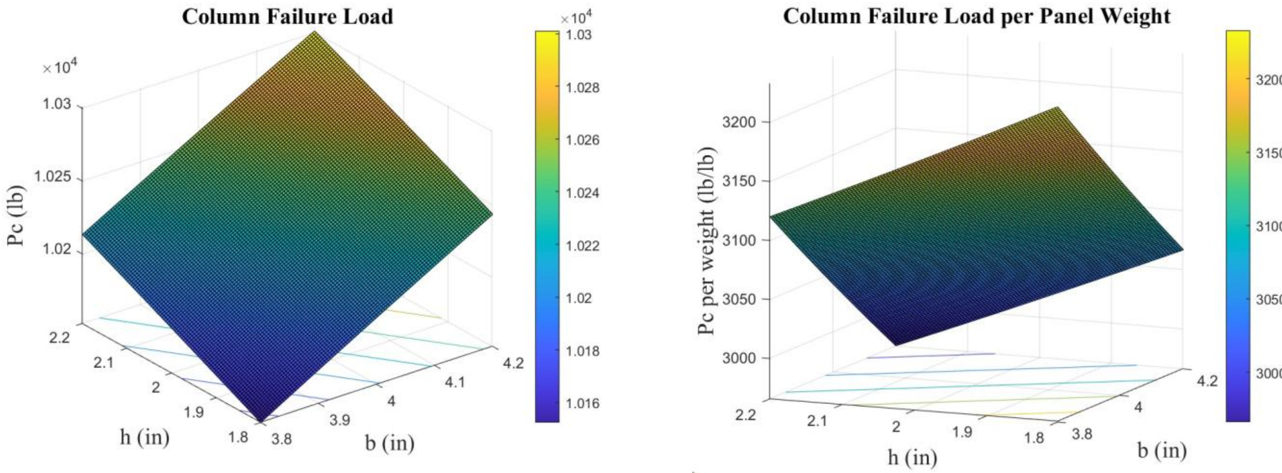

Figure 37. Hat-Panel ~~Graph~~ Optimization ~~Results~~Graph

```
=== For the maximum load only ===
Pc_max=10301.01 lb
h at Pc_max=2.20 in
b at Pc_max=4.20 in
=== For the maximum load per panel weight ===
Pc per weight_max=3120.13 lb/lb
h at Pc per weight_max=1.80 in
b at Pc per weight_max=3.80 in
```

Figure 38. Hat-Panel Optimization Results

Appendix D. Abaqus Modeling and Simulation Details

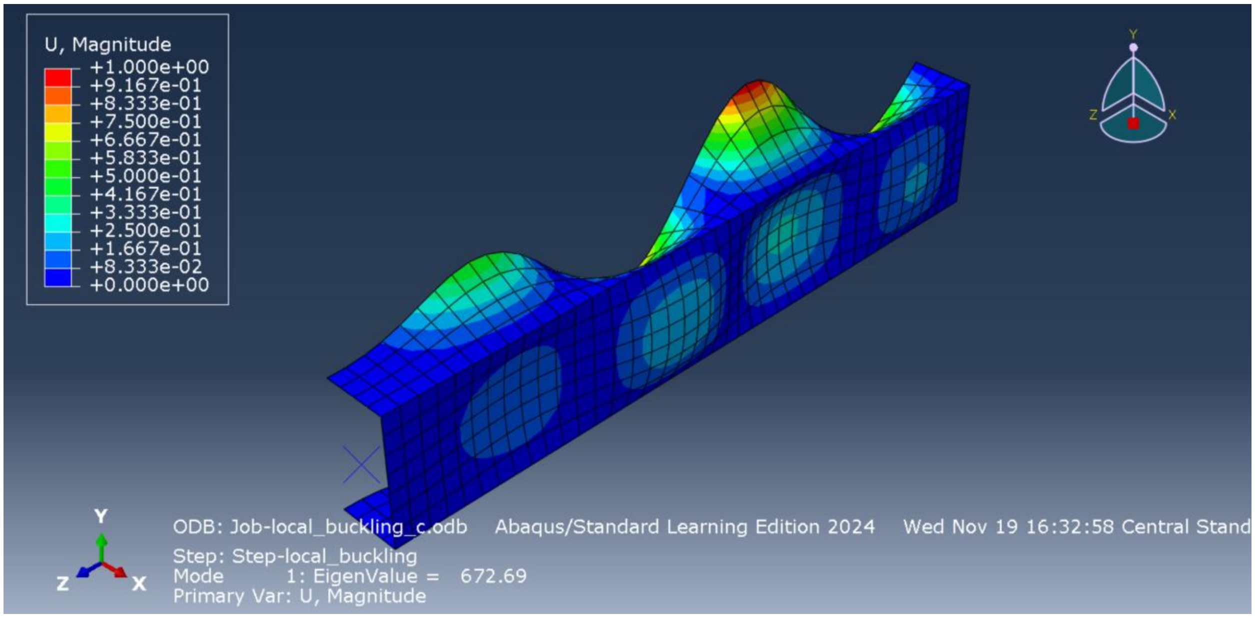

Figure 39. C-Stringer – Local Buckling Stress

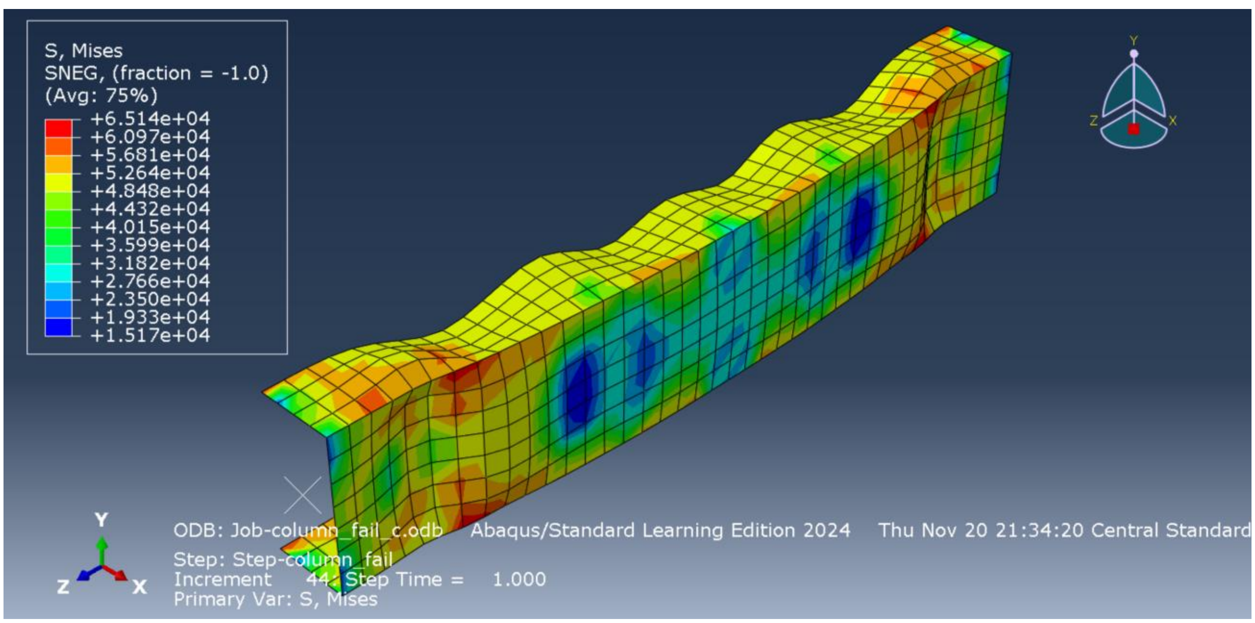

Figure 40. C-Stringer – Column Failing Stress

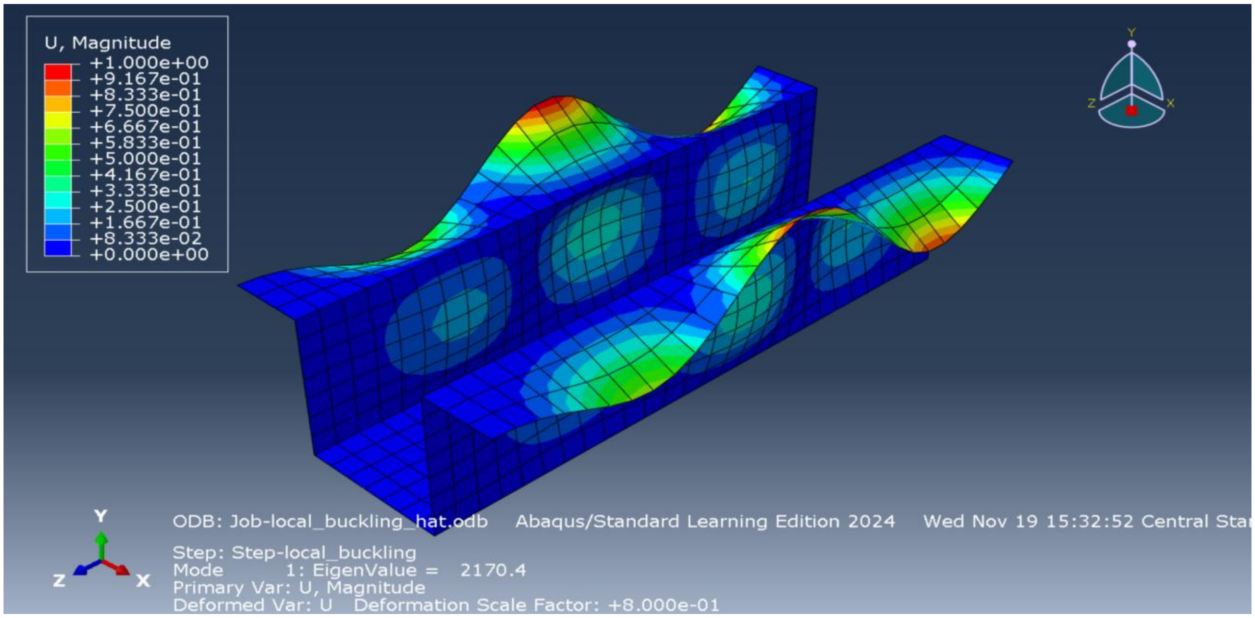

Figure 41. Hat-Stringer – Local Buckling Stress

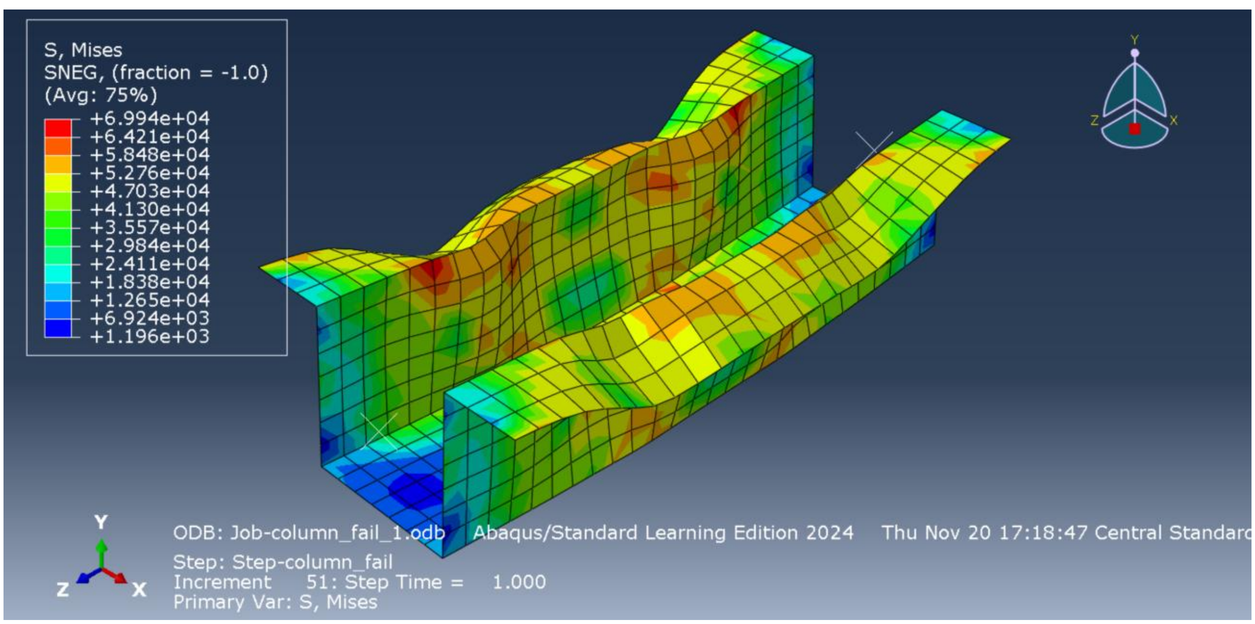

Figure 42. Hat-Stringer – Column Failing Stress